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Electrical Service Design Project

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Project Overview

Building Specifications

- 14-storied residential building
- 4 units per floor
- Each unit contains: Bedroom-1, Bedroom-2, Kitchen, Bathroom-1,Bathroom-2, Dining Room, Veranda-1, Veranda-2
- Additional Ground Floor and Basement

Project Objectives

The primary objective is to design the complete electrical services of a 4-unit, 14-storied residential building, including:

- Electrical fitting and fixtures layout diagram
- Electrical conduit diagram and schedule of conduits
- Electrical conduit diagram for power sockets
- Switchboard diagrams
- Distribution boards (SDB through MDB) diagrams (SLD)
- Substation design: layout (power flow) diagram and SLD
- Bonding/Grounding/Earthing system design
- Lightning Protection System (LPS) design and earthing for LPS
- All relevant calculations and necessary drawings

Legends and Symbols

Fitting Name	Symbol
LED Tube Light	
LED Bulb	
Ceiling Light	
Ceiling Fan	
Exhaust Fan	
2 pin Socket	

Figure: Legends and Symbols — Part 1

Fitting Name and Description	Symbol
3 Pin Socket	
Switch Board	
Emergency Switch Board	
Sub Distribution Board	
Emergency Sub Distribution Board	
Main Distribution Board	
Emergency Main Distribution Board (EMDB)	

Figure: Legends and Symbols — Part 2

Unit Floor Layout

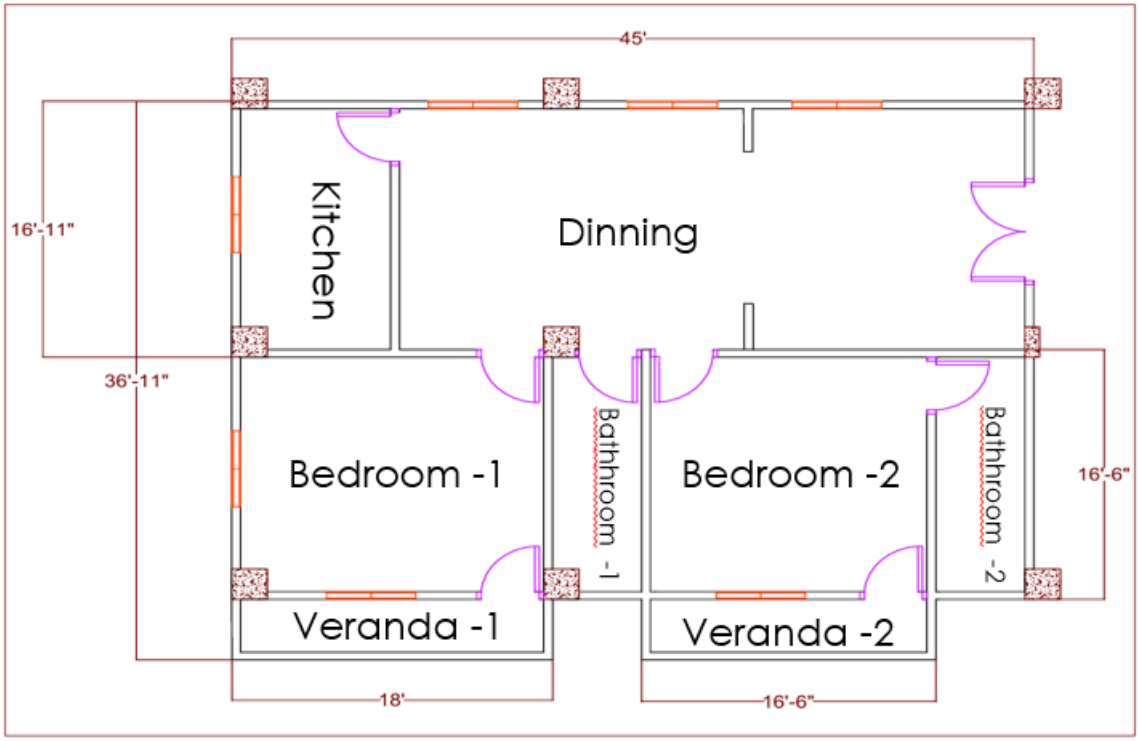


Figure: Typical Unit Floor Layout — Bedroom-1, Bedroom-2, Kitchen, Bathroom-1, Bathroom-2, Dining Room, Veranda-1, Varenda-2

Fixture Calculations

Formula and Methodology

The number of lights required for each room is calculated using the lumen method formula:

$$N = (E \times L \times W) / (F \times UF \times MF)$$

Where:

- N = Number of lights required
- E = Required illuminance level (lux) per BNBC2020 Table 8.1.5
- L = Room length (m), W = Room width (m)
- F = Average luminous flux per lamp (lumen). Assumed 2250 lm at 20W for all rooms; 3200 lm at 30W for Dining Space only
- UF = Utilization factor (depends on Room Index and surface reflectances)
- MF = Maintenance factor = 0.8 (20% deterioration assumed due to dust and aging)

Room Index is calculated as:

$$Room\ Index = (L \times W) / (Mounting\ Height \times (L + W))$$

The number of fans required is calculated as:

$$N_fans = (L\ (ft) \times W\ (ft)) / 100$$

Fixture Calculation Table

Area or Activity	Illuminance (lux)
Dwelling Houses	
Bedrooms	
General	70
Bed-head, Dressing table	250
Kitchens	200
Dining rooms (tables)	150
Bathrooms	
General	100
Shaving, make-up	300
Stairs	100
Lounges	100
Garages & Porches	100
Basement Car Park	100
Porches, Entrances	70
Sewing and darning	600
Reading (casual)	150
Home work and sustained reading	300

Figure: Unit Room Fixture Calculation Table

Chart for Calculating Utilization Factor:

Room reflectances			Room index								
C	W	F	0.75	1.00	1.25	1.50	2.00	2.50	3.00	4.00	5.00
0.7	0.5	0.2	NA	0.61	0.65	0.67	0.70	0.71	0.73	0.74	0.75

- **Room Index Formula:**

$$RoomIndex = \frac{L*W}{MountingHeight*(L+W)}$$

- **Total Lumen Formula:**

$$TotalLumen = \sum N_i * F_i = \frac{E*L*W}{UF*LLF}$$

Figure: Fixture Calculation — Illuminance and Lumen Data

- **Number of Fans Formula:**

$$Numberof Fans = \frac{L*W(insq.feet)}{100}$$

Figure: Fixture Calculation — Room Index and UF Reference

Calculations:

The following elaborated calculations present the step-by-step lumen method for each room in the typical unit. All dimensions are converted from feet-inches to SI units. Reflectance values: ceiling = 0.7, wall = 0.5, floor = 0.2. Utilization Factors (UF) are obtained from the manufacturer's chart using the computed Room Index.

〈 For Bedroom-1

Length = 15'6" = 4.724 m
Width = 17'0" = 5.182 m
Area, A = 4.724 × 5.182 = 24.48 m²
Luminaire Height = 9 ft
Working Plane Height = 3 ft
Mounting Height (MH) = 9 − 3 = 6 ft = 1.829 m
Room Index = (L × W) / (MH × (L + W)) = (4.724 × 5.182) / (1.829 × (4.724 + 5.182)) = 1.3513
Utilization Factor, UF = 0.57 (from UF chart for Room Index = 1.3513)
Maintenance Factor, MF = 0.8
Required Illuminance, E = 70 lux (Bedroom, per BNBC 2020 Table 8.1.5)
Luminous Flux per lamp, F = 2250 lm [lm (20 W fluorescent tubelight)]
Number of lights, N = (E × L × W) / (F × UF × MF)
N = (70 × 4.724 × 5.182) / (2250 × 0.57 × 0.8) = 1,713.6 / 1,026 = 1.6702
Calculated N = 1.6702

➤ ***We will install 2 tubelights.***

Number of Fans = (L × W in sq. ft.) / 100
Number of Fans = (15.5 × 17.0) / 100 = (15.5 × 17.0) / 100 = 263.50 / 100 = 2.6350

➤ ***We will install 2 ceiling fans.***

〈 For Bedroom-2

Length = 15'6" = 4.724 m
Width = 15'6" = 4.724 m
Area, A = 4.724 × 4.724 = 22.32 m²

Luminaire Height = 9 ft
Working Plane Height = 3 ft
Mounting Height (MH) = 9 − 3 = 6 ft = 1.829 m
Room Index = (L × W) / (MH × (L + W)) = (4.724 × 4.724) / (1.829 × (4.724 + 4.724)) = 1.2917
Utilization Factor, UF = 0.552 (from UF chart for Room Index = 1.2917)
Maintenance Factor, MF = 0.8
Required Illuminance, E = 70 lux (Bedroom, per BNBC 2020 Table 8.1.5)
Luminous Flux per lamp, F = 2250 lm [lm (20 W fluorescent tubelight)]
Number of lights, N = (E × L × W) / (F × UF × MF)
N = (70 × 4.724 × 4.724) / (2250 × 0.552 × 0.8) = 1,562.4 / 993.6 = 1.5725
Calculated N = 1.5725

➤ ***We will install 2 tubelights.***

Number of Fans = (L × W in sq. ft.) / 100
Number of Fans = (15.5 × 15.5) / 100 = (15.5 × 15.5) / 100 = 240.25 / 100 = 2.4025

➤ ***We will install 2 ceiling fans.***

〈 For Bathroom-1 & Bathroom-2 (each)

Length = 15'6" = 4.724 m
Width = 5'0" = 1.524 m
Area, A = 4.724 × 1.524 = 7.20 m²
Luminaire Height = 7 ft
Working Plane Height = 3 ft
Mounting Height (MH) = 7 − 3 = 4 ft = 1.219 m
Room Index = (L × W) / (MH × (L + W)) = (4.724 × 1.524) / (1.219 × (4.724 + 1.524)) = 0.9451
Utilization Factor, UF = 0.4756 (from UF chart for Room Index = 0.9451)
Maintenance Factor, MF = 0.8
Required Illuminance, E = 100 lux (Bathroom, per BNBC 2020 Table 8.1.5)
Luminous Flux per lamp, F = 2250 lm [lm (20 W fluorescent lamp)]
Number of lights, N = (E × L × W) / (F × UF × MF)
N = (100 × 4.724 × 1.524) / (2250 × 0.4756 × 0.8) = 719.9 / 856.1 = 0.8410
Calculated N = 0.8410

➤ ***We will install 1 light per bathroom. (Ceiling fans are not installed in bathrooms.)***

〈 For Dining Room

Length = 35'1" = 10.693 m
Width = 15'11" = 4.852 m
Area, A = 10.693 × 4.852 = 51.88 m²
Luminaire Height = 9 ft
Working Plane Height = 3.5 ft
Mounting Height (MH) = 9 − 3.5 = 5.5 ft = 1.676 m
Room Index = (L × W) / (MH × (L + W)) = (10.693 × 4.852) / (1.676 × (10.693 + 4.852)) = 1.9908
Utilization Factor, UF = 0.66 (from UF chart for Room Index = 1.9908)
Maintenance Factor, MF = 0.8
Required Illuminance, E = 150 lux (Dining Room, per BNBC 2020 Table 8.1.5)
Luminous Flux per lamp, F = 3200 lm [lm (30 W fluorescent tubelight — higher flux for dining space)]
Number of lights, N = (E × L × W) / (F × UF × MF)

N = (150 × 10.693 × 4.852) / (3200 × 0.66 × 0.8) = 7,782.0 / 1,689.6 = 4.6056
Calculated N = 4.6056

➤ ***We will install 4 tubelights (rounded down; sufficient for required illuminance).***

Number of Fans = (L × W in sq. ft.) / 100
Number of Fans = (35.083 × 15.917) / 100 = (35.083 × 15.917) / 100 = 558.23 / 100 = 5.5823

➤ ***We will install 3 ceiling fans (reduced for economy; adequate air circulation maintained).***

〈 For Kitchen

Length = 15'11" = 4.852 m
Width = 8'5" = 2.566 m
Area, A = 4.852 × 2.566 = 12.45 m²
Luminaire Height = 9 ft
Working Plane Height = 4 ft
Mounting Height (MH) = 9 − 4 = 5 ft = 1.524 m
Room Index = (L × W) / (MH × (L + W)) = (4.852 × 2.566) / (1.524 × (4.852 + 2.566)) = 1.1021
Utilization Factor, UF = 0.514 (from UF chart for Room Index = 1.1021)
Maintenance Factor, MF = 0.8
Required Illuminance, E = 200 lux (Kitchen/task area, per BNBC 2020 Table 8.1.5)
Luminous Flux per lamp, F = 2250 lm [lm (20 W fluorescent tubelight)]
Number of lights, N = (E × L × W) / (F × UF × MF)
N = (200 × 4.852 × 2.566) / (2250 × 0.514 × 0.8) = 2,490.0 / 924.0 = 2.6904
Calculated N = 2.6904

➤ ***We will install 2 tubelights.***

Number of Fans = (L × W in sq. ft.) / 100
Number of Fans = (15.917 × 8.417) / 100 = (15.917 × 8.417) / 100 = 133.97 / 100 = 1.3397

➤ ***We will install 1 ceiling fan.***

〈 For Veranda-1

Length = 17'0" = 5.182 m
Width = 3'6" = 1.067 m
Area, A = 5.182 × 1.067 = 5.53 m²
Luminaire Height = 9 ft
Working Plane Height = 3 ft
Mounting Height (MH) = 9 − 3 = 6 ft = 1.829 m
Room Index = (L × W) / (MH × (L + W)) = (5.182 × 1.067) / (1.829 × (5.182 + 1.067)) = 0.4837
Utilization Factor, UF = 0.35 (from UF chart for Room Index = 0.4837)
Maintenance Factor, MF = 0.8
Required Illuminance, E = 70 lux (Veranda/semi-outdoor, per BNBC 2020 Table 8.1.5)
Luminous Flux per lamp, F = 2250 lm [lm (20 W fluorescent lamp)]
Number of lights, N = (E × L × W) / (F × UF × MF)
N = (70 × 5.182 × 1.067) / (2250 × 0.35 × 0.8) = 386.9 / 630.0 = 0.6142
Calculated N = 0.6142

➤ ***We will install 1 light. (No ceiling fan — open veranda space.)***

〈 For Veranda-2

Length = 15'6" = 4.724 m
Width = 3'6" = 1.067 m
Area, A = 4.724 × 1.067 = 5.04 m²
Luminaire Height = 9 ft
Working Plane Height = 3 ft
Mounting Height (MH) = 9 − 3 = 6 ft = 1.829 m
Room Index = (L × W) / (MH × (L + W)) = (4.724 × 1.067) / (1.829 × (4.724 + 1.067)) = 0.4759
Utilization Factor, UF = 0.35 (from UF chart for Room Index = 0.4759)
Maintenance Factor, MF = 0.8
Required Illuminance, E = 70 lux (Veranda/semi-outdoor, per BNBC 2020 Table 8.1.5)
Luminous Flux per lamp, F = 2250 lm [lm (20 W fluorescent lamp)]
Number of lights, N = (E × L × W) / (F × UF × MF)
N = (70 × 4.724 × 1.067) / (2250 × 0.35 × 0.8) = 352.8 / 630.0 = 0.5540
Calculated N = 0.5540

➤ ***We will install 1 light. (No ceiling fan — open veranda space.)***

Schedule of Conduit

Symbol	Wire Rating (single core)-mm²	Current Rating (ampere)	GI Pipe Diameter (inch)
C1	2x1.5	5A	¾
C2	4x1.5	5A	¾
C3	6x1.5	5A	¾
C4	8x1.5	5A	¾
C5	10x1.5	5A	1
C6	12x1.5	5A	1
C7	14x1.5	5A	1
C8	2x4+4 ECC	15A	1
C9	2x6+6 ECC	20A	1

Figure: Unit Floor — General Conduit (Detail)

Complete Floor Layout

The complete floor layout shows all four units arranged on a single floor, along with the common areas.

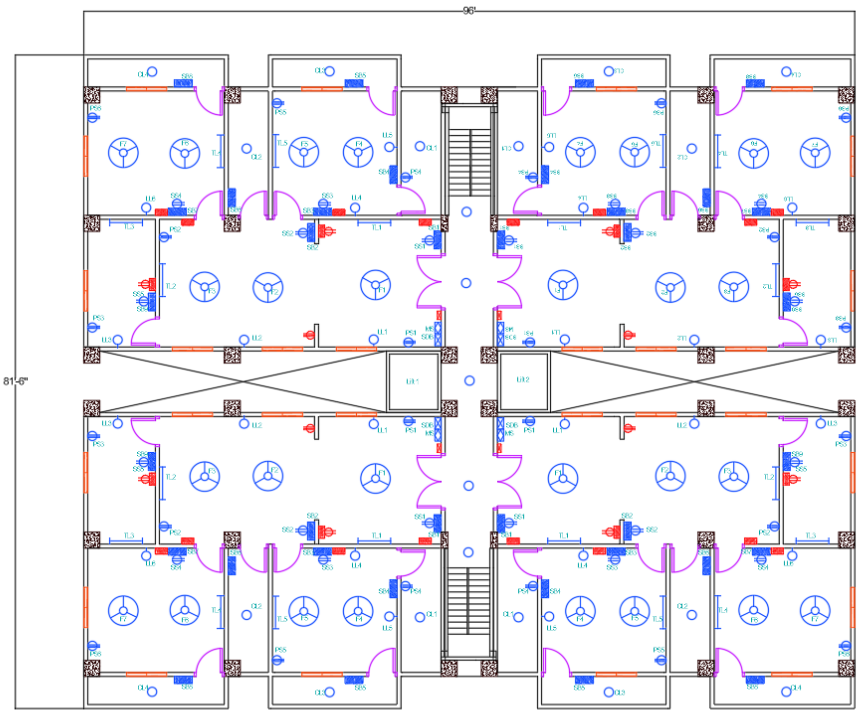


Figure: Complete Floor Layout — All 4 Units

Unit Design

Unit Floor — Fittings and Fixtures

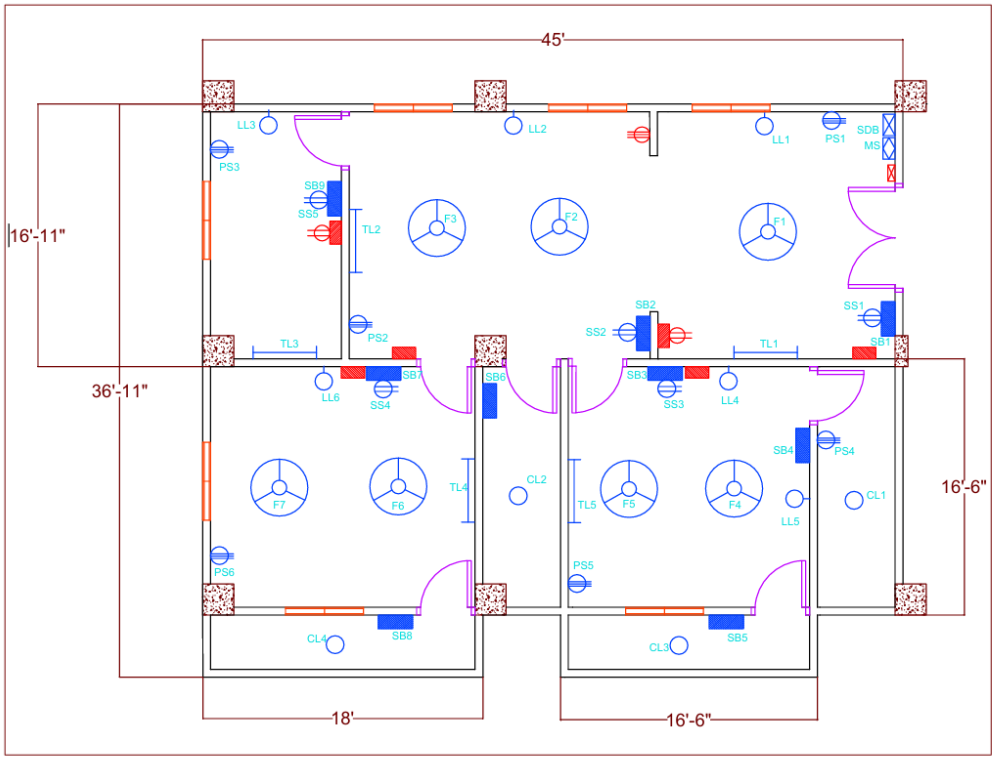


Figure: Unit Floor — Fittings and Fixtures Layout

Unit Floor — General Conduit Layout

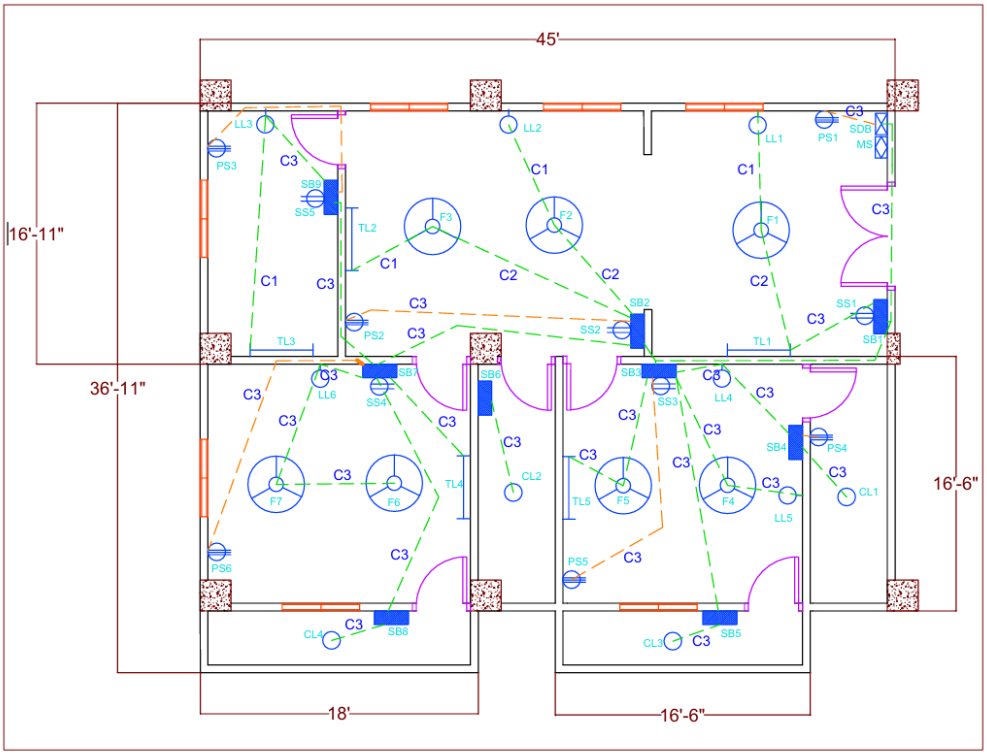


Figure: Unit Floor — General Conduit Layout

Unit Floor — Emergency Conduit Layout

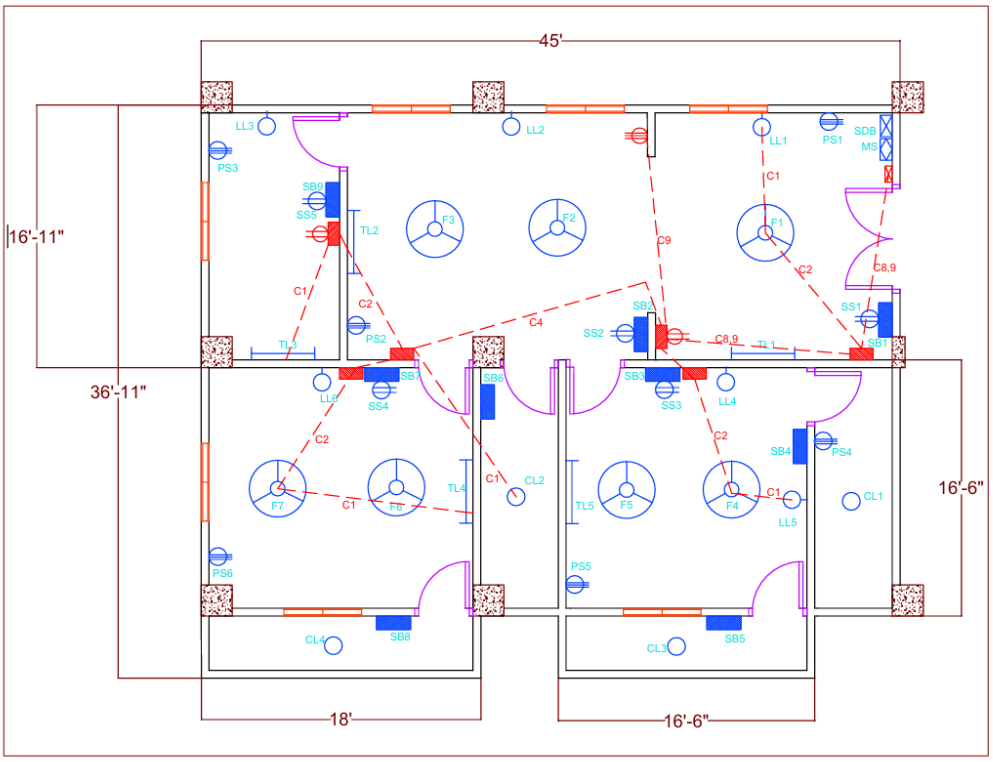


Figure: Unit Floor — Emergency Conduit Layout

Unit Floor — Power Socket Layout

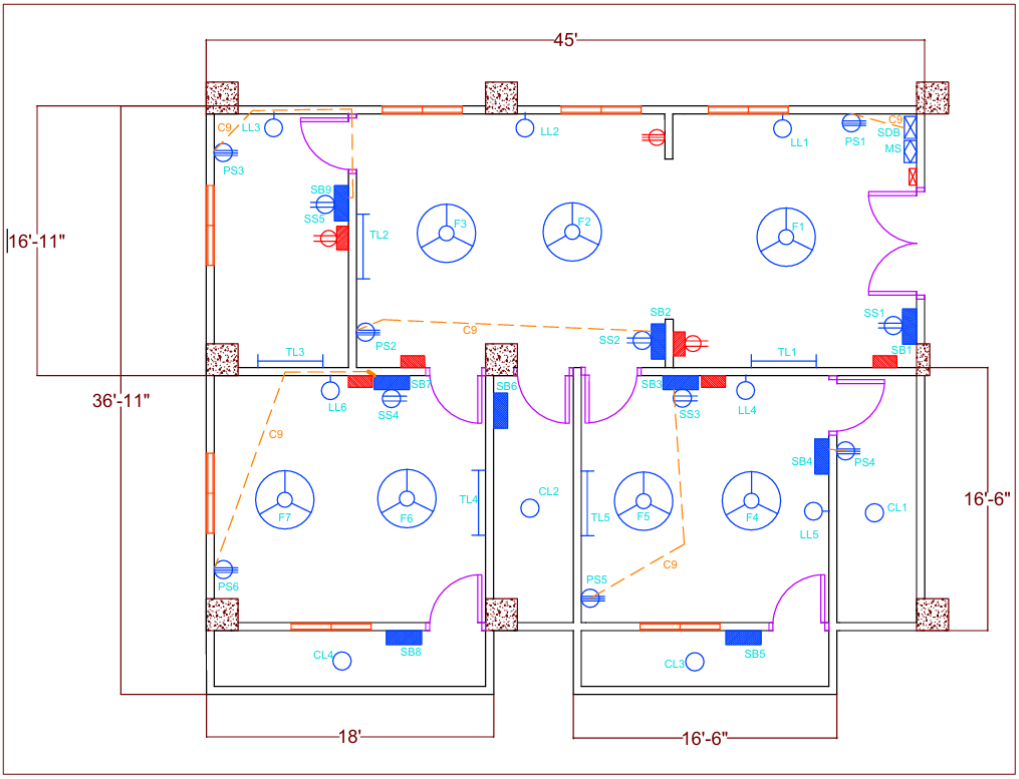


Figure: Unit Floor — Power Socket Layout

Unit Floor — All Conduit Combined

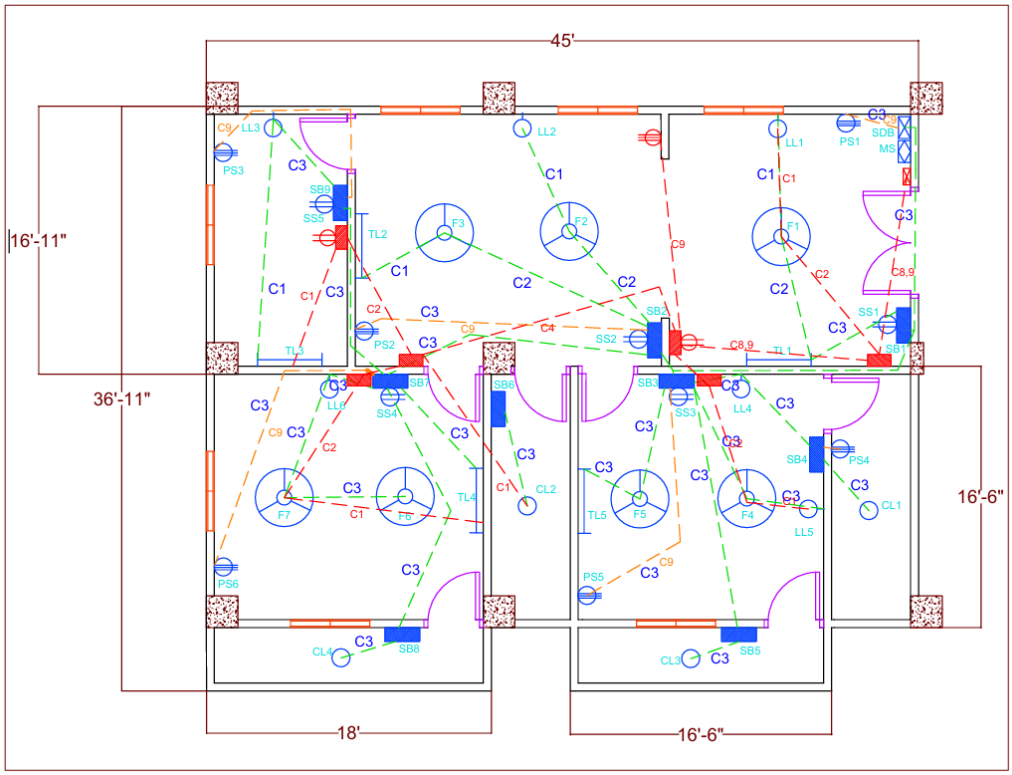


Figure: Unit Floor — All Conduit Combined Layout

Switchboard and Distribution Board Diagrams

Switchboard Diagram — Calculation

The switchboard diagram is calculated based on the actual load connected to each circuit. Current is calculated using I = P / (V × PF) where PF = 0.6 for lights and 0.8 for fans and sockets.

Room Name	Circuit No	Switch Board	Fixture	Power(W)	Current Rating	Breaker to SDB
Bed 01	Circuit 01	SB3	F5	70	0.32	
			TL5	20	0.1	
			LL4	20	0.1	
			F4	70	0.32	
			SS3		5	
			LL5	20	0.1	
		SB4	CL1	20	0.1	
		SB5	CL3	20	0.1	
			Total		**6.14 A**	10A
Bed 02	Circuit 02	SB7	SS4		5	
			LL6	20	0.1	
			F7	70	0.32	
			F6	70	0.32	
			TL4	20	0.1	
		SB8	CL4	20	0.1	
			Total		**5.94 A**	10A
Dinning Room	Circuit 03	SB1	SS1		5	
			TL1	30	0.15	
			F1	70	0.32	
			LL1	20	0.1	
		SB2	SS2		5	
			F2	70	0.32	
			LL2	20	0.1	
			TL2	30	0.15	
		SB6	CL2	30	0.15	
			Total		**11.29 A**	15A
Kitchen	Circuit 04	SB9	SS5		5	
			LL3	20	0.1	
			TL3	20	0.1	
			Total		**5.2 A**	10A

Figure: Switchboard Diagram Calculation Table

Name	Power Scoket	Current Rating	Conduit	Conduit Diameter
Bed 01	PS4	20	C9	25mm
	PS5	15	C9	25mm
Bed 02	PS6	15	C9	25mm
Dinning Room	PS1	15	C9	25mm
	PS2	20	C9	25mm
Kitchen	PS3	20	C9	25mm

Figure: Switchboard Calculation — Current and Wire Rating

Switchboard Diagram (SB Connection)

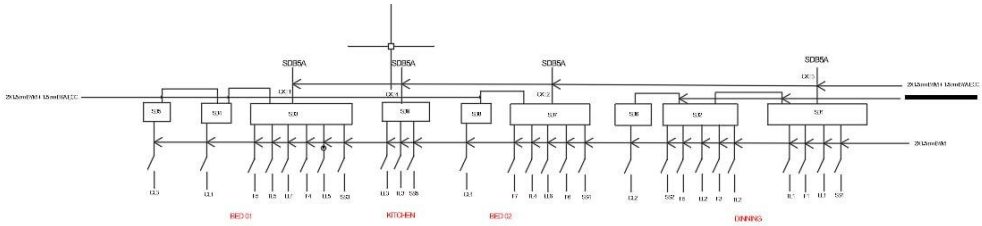


Figure: Switch Board Connection Diagram

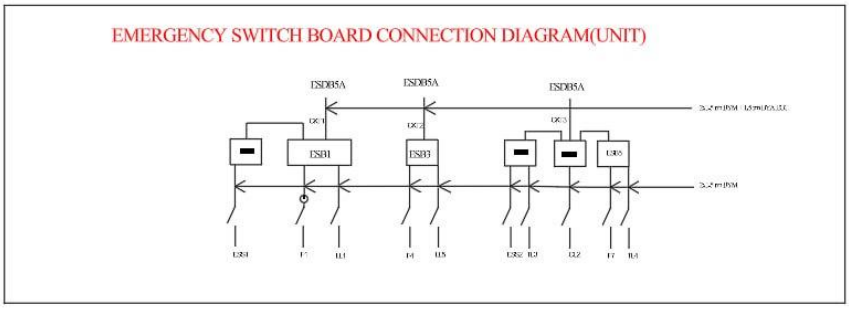


Figure: Switch Board MCB Connection Diagram

Sub Distribution Board (SDB) Diagram

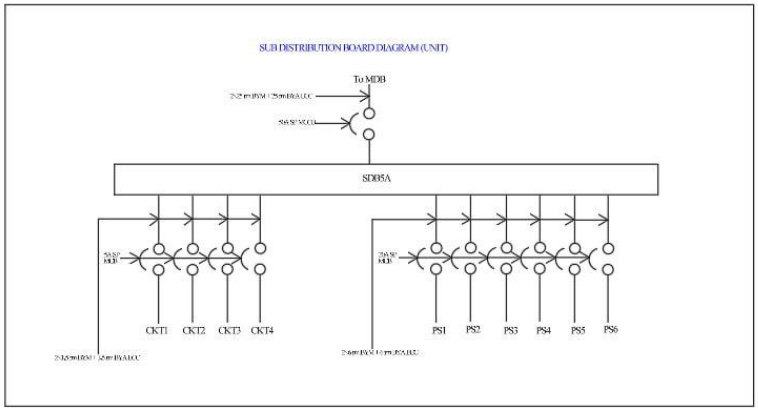


Figure: Sub Distribution Board (SDB) Diagram

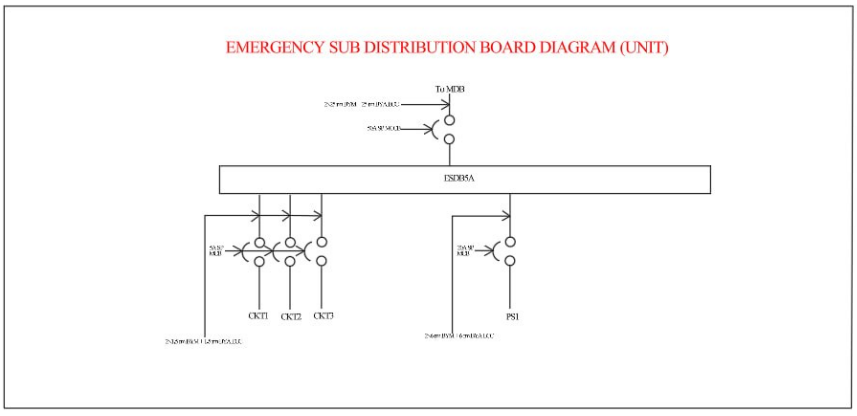


Figure: SDB — MCB and Wire Schedule

Room Name	Circuit No	Switch Board	Fixture	Power(W)	Current Rating	Breaker to SDB
Bed 01	Circuit 01	SB3	F5	70	0.4	
			TL5	20	0.1	
			LL4	20	0.1	
			F4	70	0.4	
			SS3		5	
			LL5	20	0.1	
		SB4	CL1	20	0.1	
		SB5	CL3	20	0.1	
			Total		**6.3A**	10A
Bed 02	Circuit 02	SB7	SS4		5	
			LL6	20	0.1	
			F7	70	0.4	
			F6	70	0.4	
			TL4	20	0.1	
		SB8	CL4	20	0.1	
			Total		**6.1A**	10A
Dinning Room	Circuit 03	SB1	SS1		5	
			TL1	30	0.15	
			F1	70	0.4	
			LL1	20	0.1	
		SB2	SS2		5	
			F2	70	0.4	
			LL2	20	0.1	
			TL2	30	0.15	
		SB6	CL2	30	0.15	
			Total		**11.45A**	15A
Kitchen	Circuit 04	SB9	SS5		5	
			LL3	20	0.1	
			TL3	20	0.1	
			Total		**5.2 A**	10A
			Total		**29.05 A**	

Figure: Unit SDB — Load Calculation Table

Name	Power Scoket	Current Rating	Conduit	Conduit Diameter
Bed 01	PS4	20	C9	25mm
	PS5	20	C9	25mm
Bed 02	PS6	20	C9	25mm
Dinning Room	PS1	20	C9	25mm
	PS2	20	C9	25mm
Kitchen	PS3	20	C9	25mm

Figure: Unit SDB — Summary Table

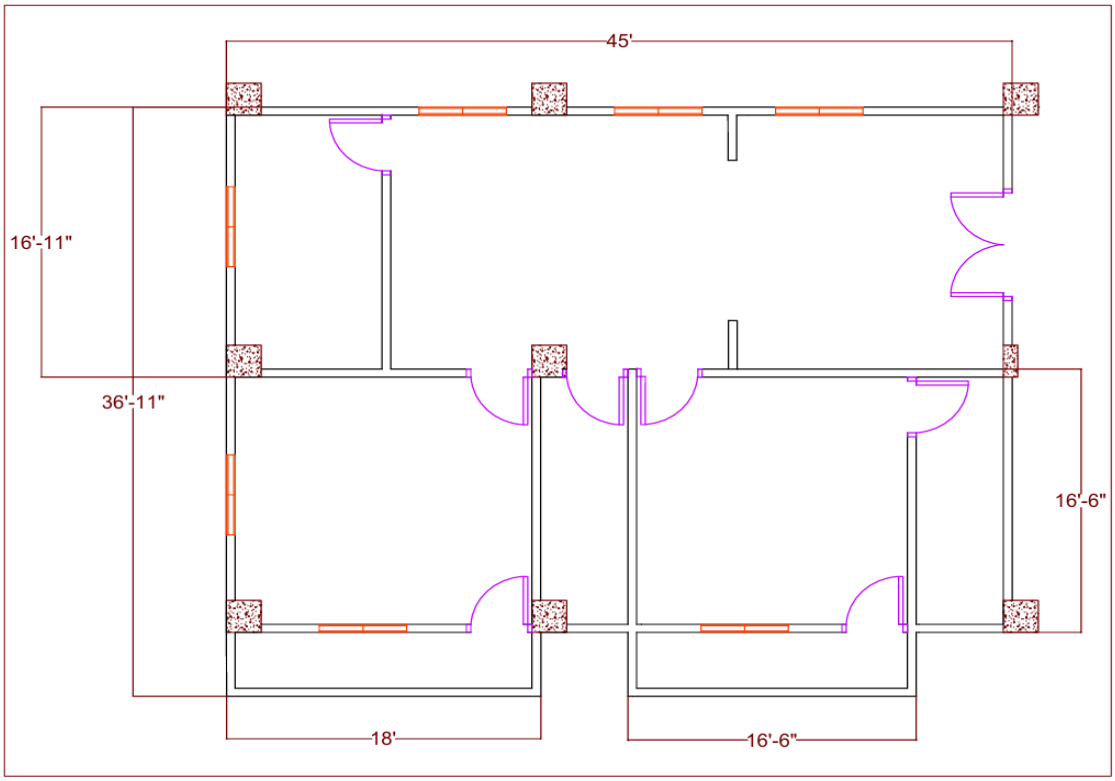


Figure: Unit Floor Plan — Reference Layout

Unit All Conduit with SDB

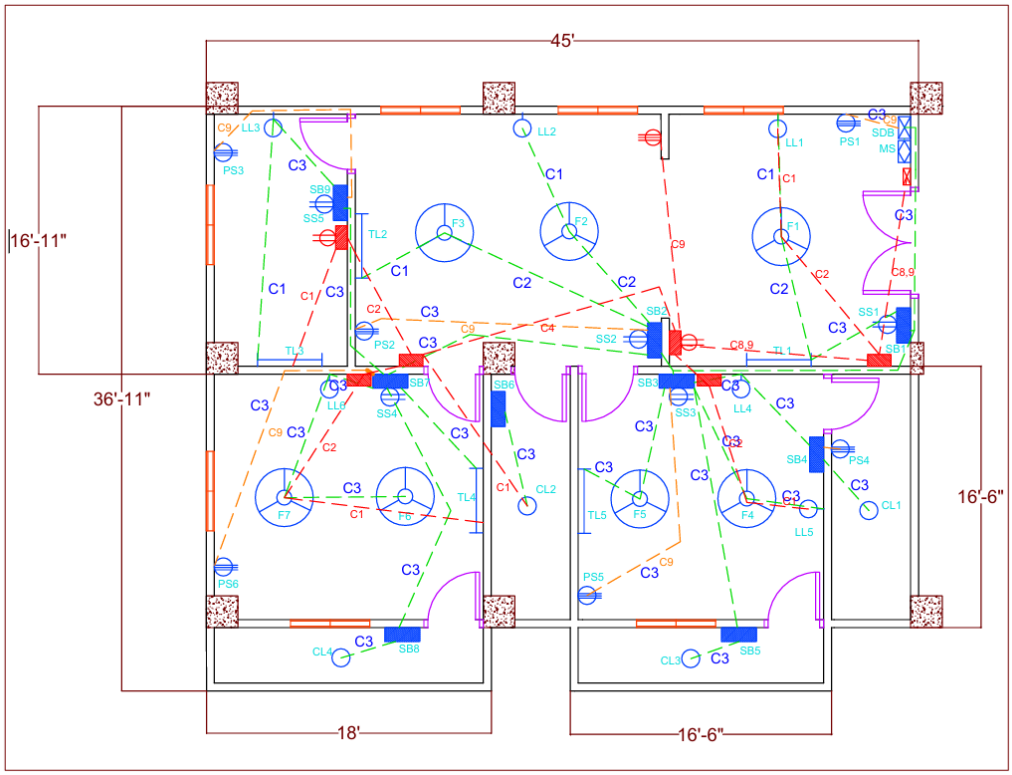


Figure: Unit — Complete Conduit Layout with SDB

Ground Floor Design

Ground Floor Layout

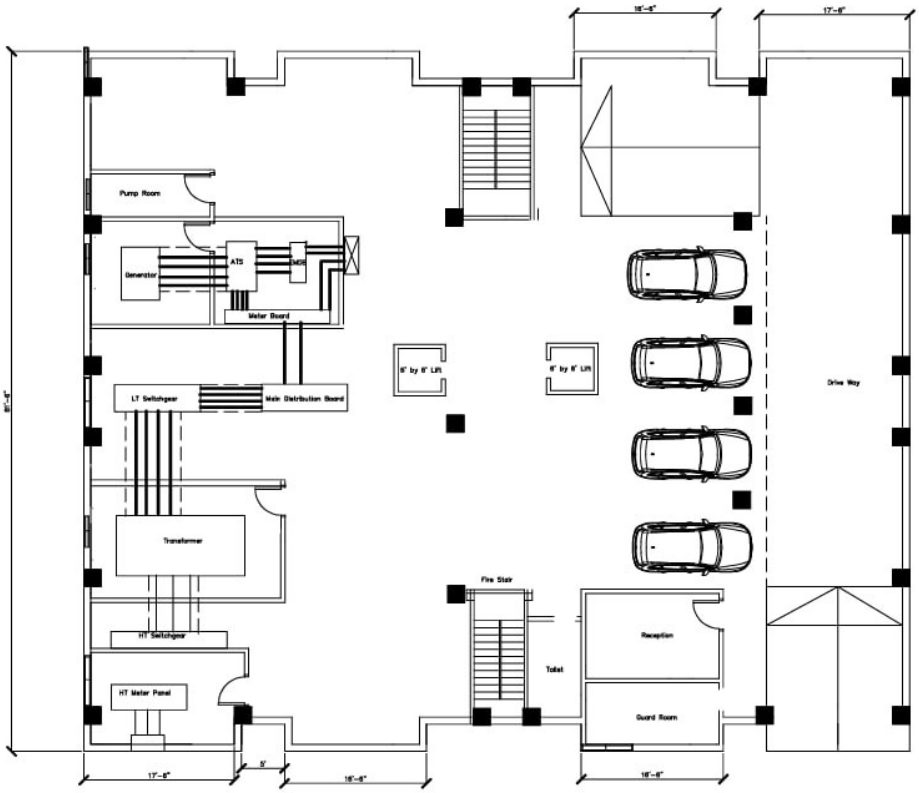


Figure: Ground Floor Layout

Ground Floor Fixture Calculations

The fixture calculation for each room of the ground floor follows the same lumen method. Key rooms calculated include the Guard Room and Reception.

Parameters	Guard Room
L(ft)	15.5
W(ft)	7
Luminaries Height	9
Working Plane Height	3
Mounting Height	6
Room Index	0.803703704
Utilization Factor	0.442
Maintenance Factor	0.8
Luminous Flux of Source	2250
Required lux	70
No. of lights	0.886876054
No. of fans	1.085

Figure: Ground Floor — Guard Room Calculation

Parameters	Guard Room
L(ft)	15.5
W(ft)	10
Luminaries Height	9
Working Plane Height	3
Mounting Height	6
Room Index	1.013071895
Utilization Factor	0.49
Maintenance Factor	0.8
Luminous Flux of Source	2250
Required lux	70
No. of lights	1.142854857
No. of fans	1.55
No. of lights used	2
No. of fans Used	2

Figure: Ground Floor — Reception Calculation

Ground Floor Fittings and Fixtures

Ground floor fittings summary:

Component	Quantity
Light	20
Fan	5
Exhaust Fan	3
Switch Board (SB)	10
Emergency SB (ESB)	5

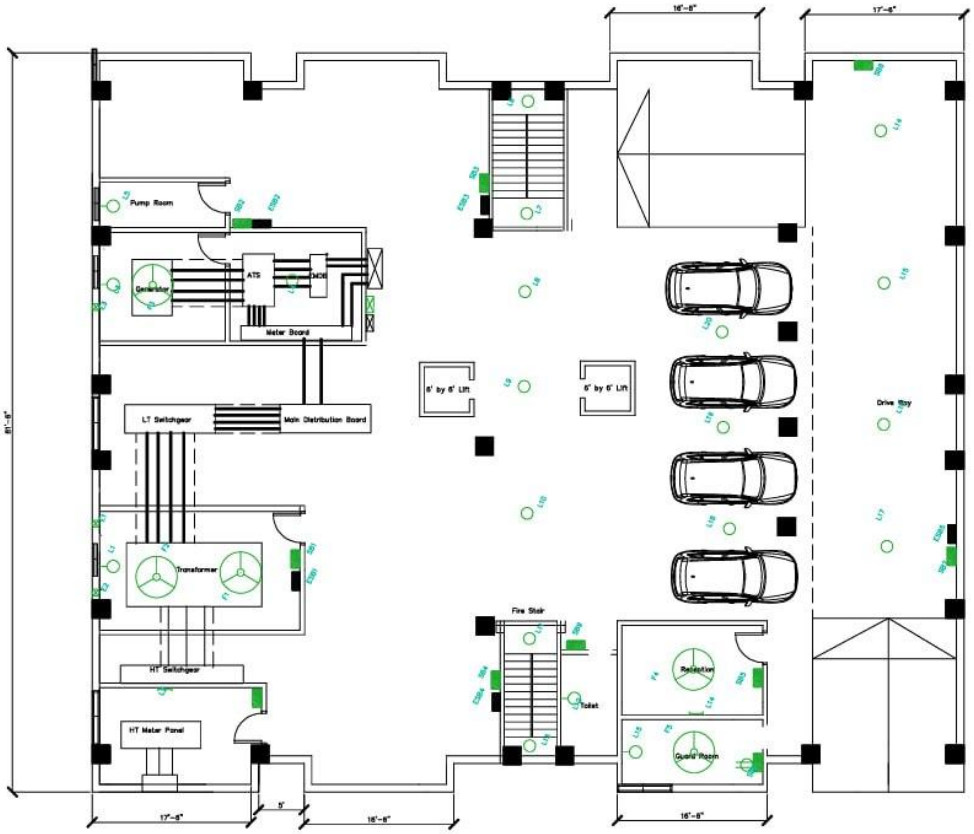


Figure: Ground Floor — Fittings and Fixtures Layout

Ground Floor — General Conduit Layout

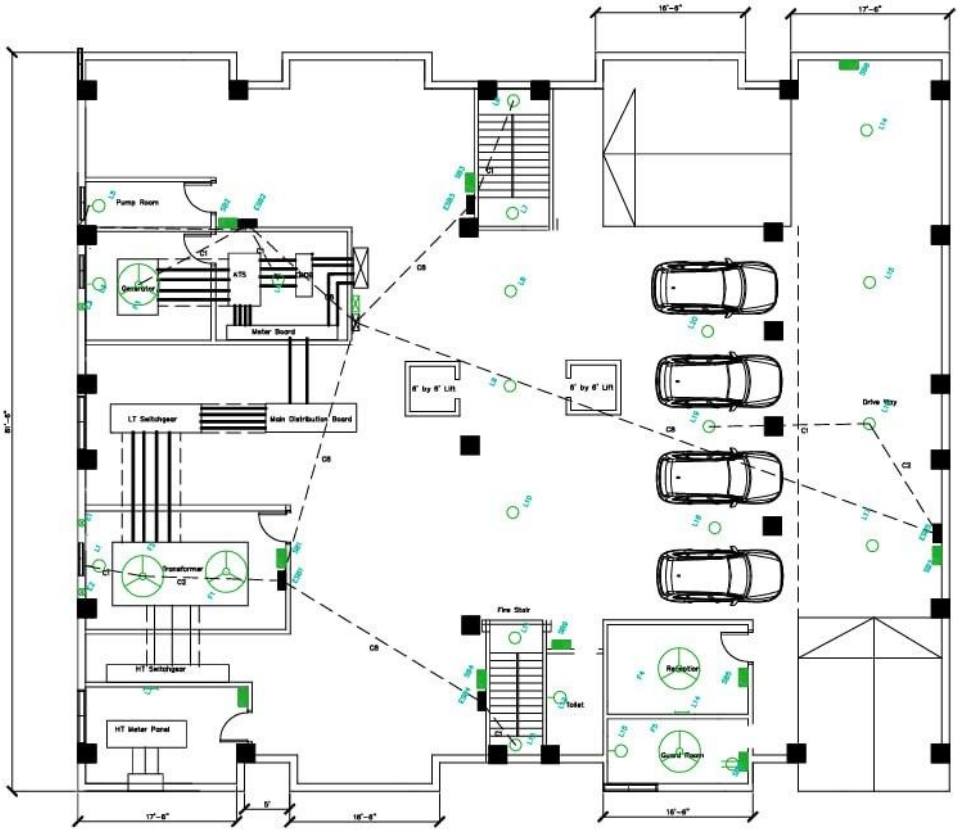


Figure: Ground Floor — General Conduit Layout

Ground Floor — Load Calculation

SDB-G load calculations

Table 1 SDB-G load calculation

to this PC

Room name	Circuit no	Switch board	fixture	Power (W)	Current rating (A)	Wire rating	Conduit diameter
Guard room	CKT2	SB6	F5 L15 SS1	80 12	0.45 0.07 5	C8	20mm
Reception	CKT2	SB5	F4 L14	80 12	0.45 0.07	C8	20mm
Transformer room	CKT2	SB1	F1 L1 E1 E2	80 12 20 20	0.45 0.07 0.12 0.12	C8	20mm
Generator room	CKT1	SB2	F3 L3 E3	80 12 20	0.45 0.07 0.12	C8	20mm
Pump room	CKT1		L5	12	0.07		
Main Stair	CKT3	SB3	L6 L7 L8	12 12 12	0.07 0.07 0.07	C8	20mm
Fire Stair	CKT2	SB4	L11 L13 L10 L9	12 12 12 12	0.07 0.07 0.07 0.07	C8	20mm
Toilet	CKT2	SB9	L12	12	0.07	C8	20mm
HT meter panel room	CKT2	SB10	L2	12	0.07	C8	20mm
Driveway and others	CKT3	SB8	L14 L15 L20	12 12 12	0.07 0.07 0.07	C8	20mm
	CKT2	SB7	L17 L18	12 12	0.07 0.07	C8	20mm

Figure: Ground Floor — Load Calculation Table

Ground Floor — Current Calculations

SDB-G Current calculations

Table 2 :SDB-G current calculations

Circuit	Location / Description	Current	Total current (A)	Breaker to SDB
CKT1	Generator room (SB2)	0.64 A	0.71	5 A
	Pump room (SB2)	0.07 A		
CKT2	Guard room (SB6)	5.52 A	7.36	10 A
	Reception (SB5)	0.52 A		
	Transformer room (SB1)	0.76 A		
	Fire Stair (SB4)	0.28 A		
	Toilet (SB9)	0.07 A		
	HT meter panel room (SB10)	0.07 A		
	Driveway and others(SB7)	0.14 A		
CKT3	Main Stair (SB3)	0.21 A	.042	5 A
	Driveway and others (SB8)	0.21 A		

Figure: Ground Floor — Current Calculation Table

Ground Floor — Emergency Conduit Layout

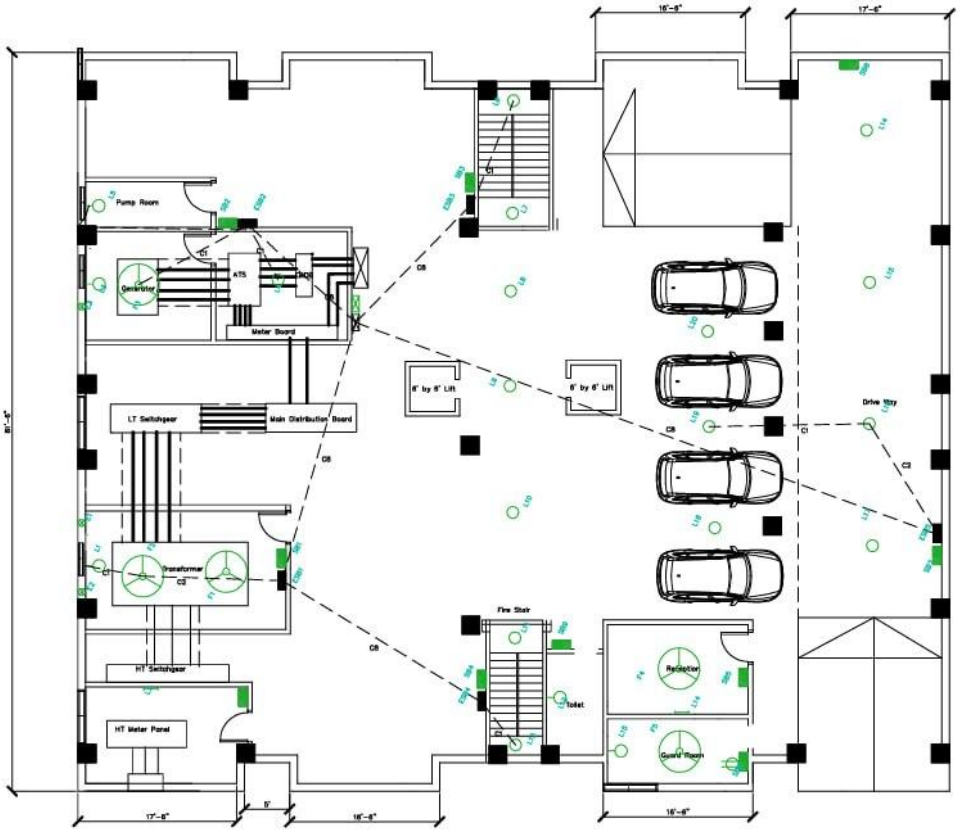


Figure: Ground Floor — Emergency Conduit Layout

Ground Floor — Emergency Load Calculation

ESDB-G load calculations

Room name	Circuit no	Switch board	fixture	Power (W)	Current rating (A)	Wire rating	Conduit diagram
Generator and pump room	CKT1	ESB2	F3 L4 L5	80 12 12	0.45 0.07 0.07	C8	20mm
Main Stair	CKT2	ESB3	L6	12	0.07	C8	20mm
Transformer room	CKT3	ESB1	F2 L1	80 12	0.45 0.07	C8	20mm
Fire Stair	CKT3	ESB4	L13	12	0.07	C8	20mm
Driveway and others	CKT4	ESB5	L16 L19	12 12	0.07 0.07	C8	20mm

Figure: Ground Floor — Emergency Load Calculation Table

Ground Floor — Emergency Current Calculation

ESDB-G current calculations

Table 4:ESDB-G current calculations

Circuit	Location / Description	Current	Total current (A)	Breaker to SDB
CKT1	Generator room (ESB2)	0.59 A	0.59	5 A
	Pump room (ESB2)			
CKT2	Main Stair (ESB3)	0.07 A	0.07	5 A
CKT3	Transformer room(ESB1)	0.52 A	0.59	5 A
	Fire Stair(ESB4)	0.07 A		
CTK 4	Driveway and others(ESB5)	0.14	0.14	5 A

Figure: Ground Floor — Emergency Current Calculation Table

Ground Floor — SDB Diagram

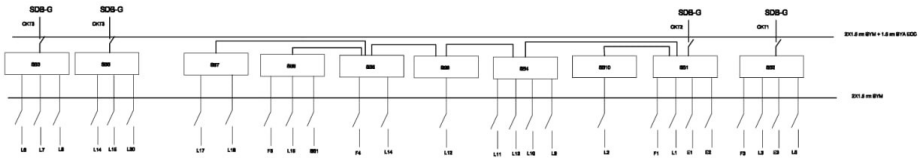


Figure: Ground Floor SDB — Circuit Diagram Part 1

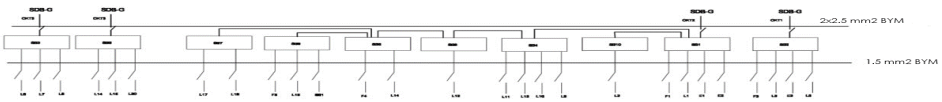


Figure: Ground Floor SDB — Wire and CB Schedule

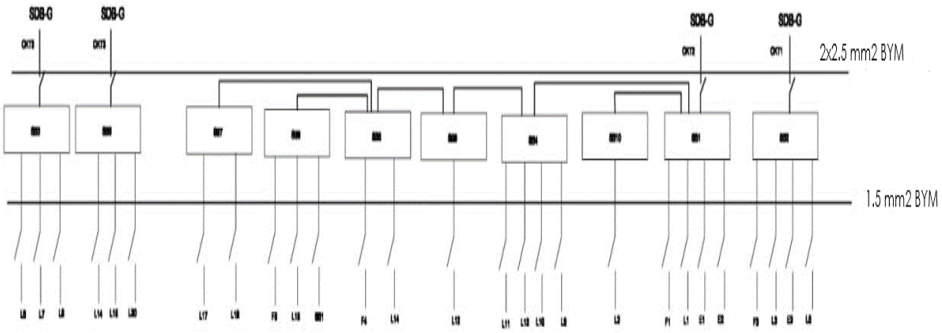


Figure: Ground Floor SDB — Diagram Part 3

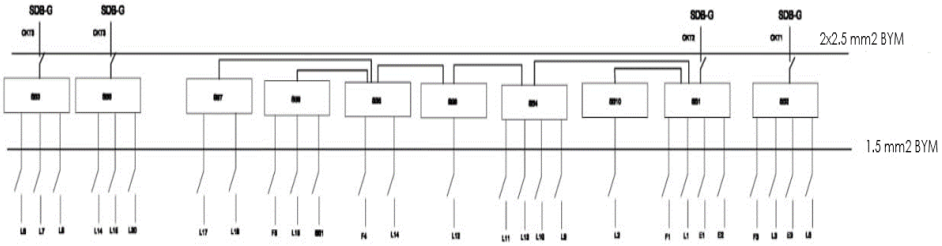


Figure: Ground Floor SDB — Complete Diagram



Figure: Ground Floor SDB — Summary

Ground Floor — ESDB Diagram

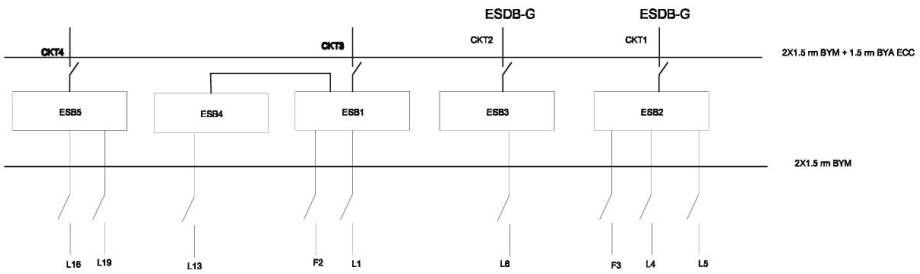


Figure: Ground Floor ESDB — Diagram Part 1

Rooftop Design

Rooftop Layout

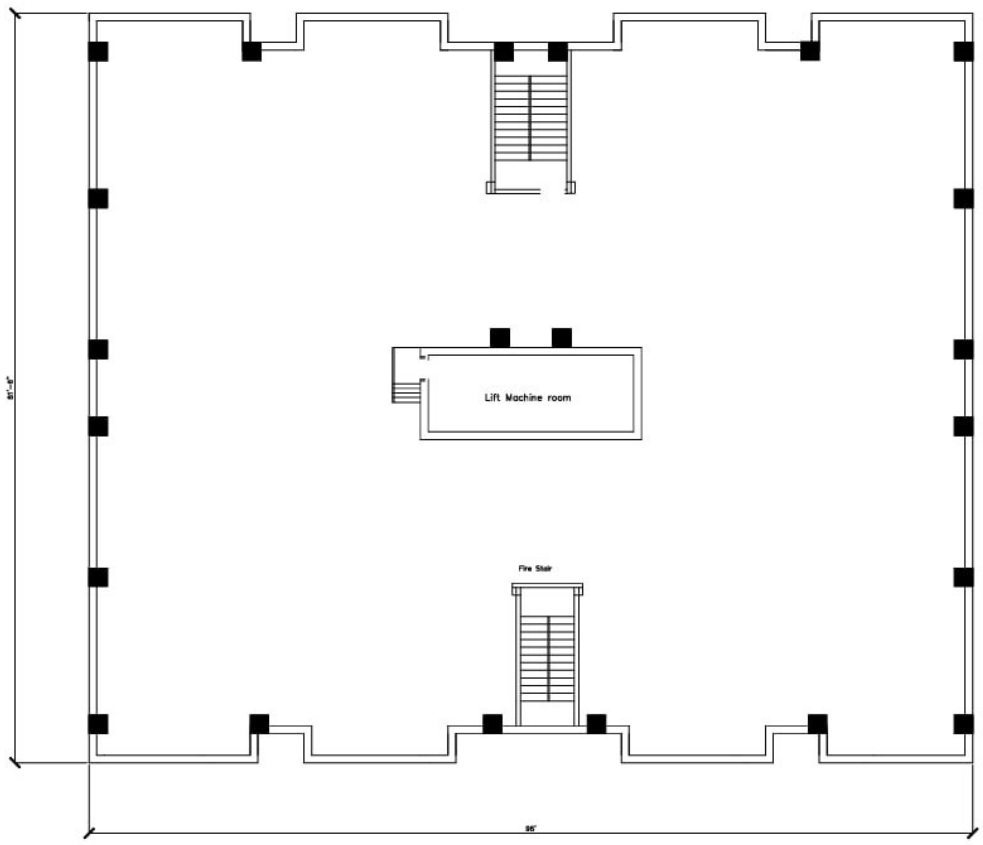


Figure: Rooftop Layout

Rooftop Fittings and Fixtures

Rooftop fittings summary:

Component	Quantity
Light	5
Switch Board (SB)	3
Emergency SB (ESB)	3

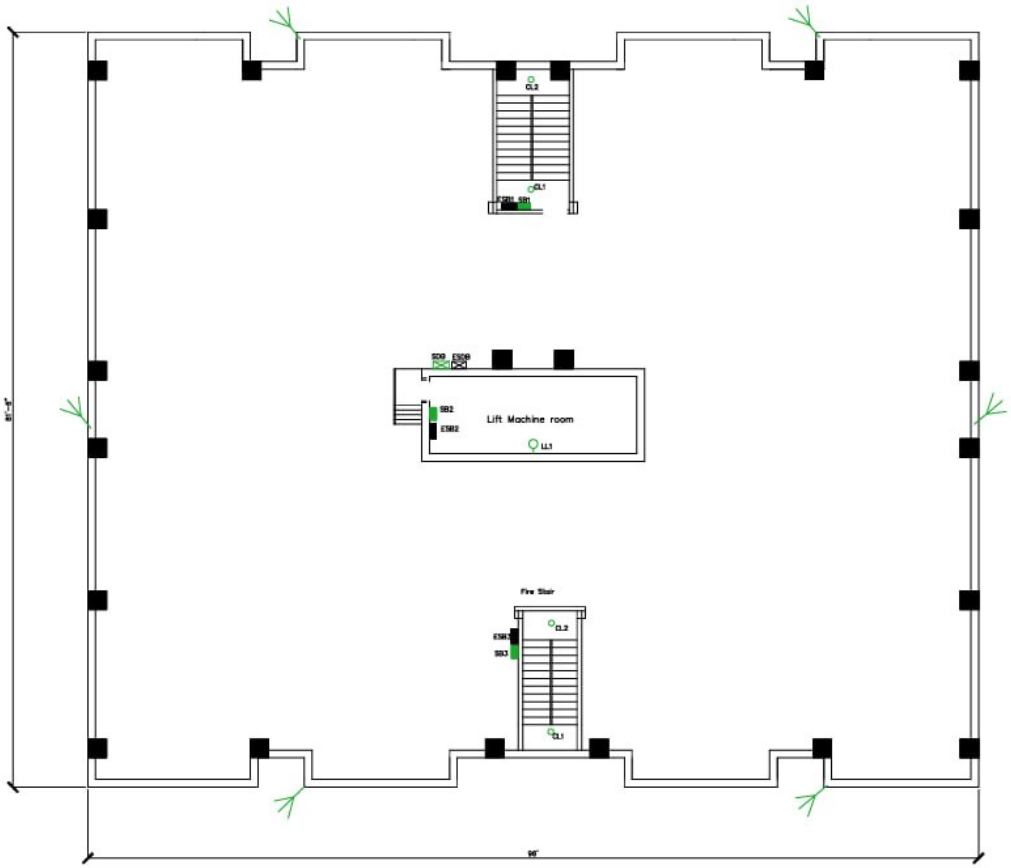


Figure: Rooftop — Fittings and Fixtures Layout

Rooftop — General Conduit Layout

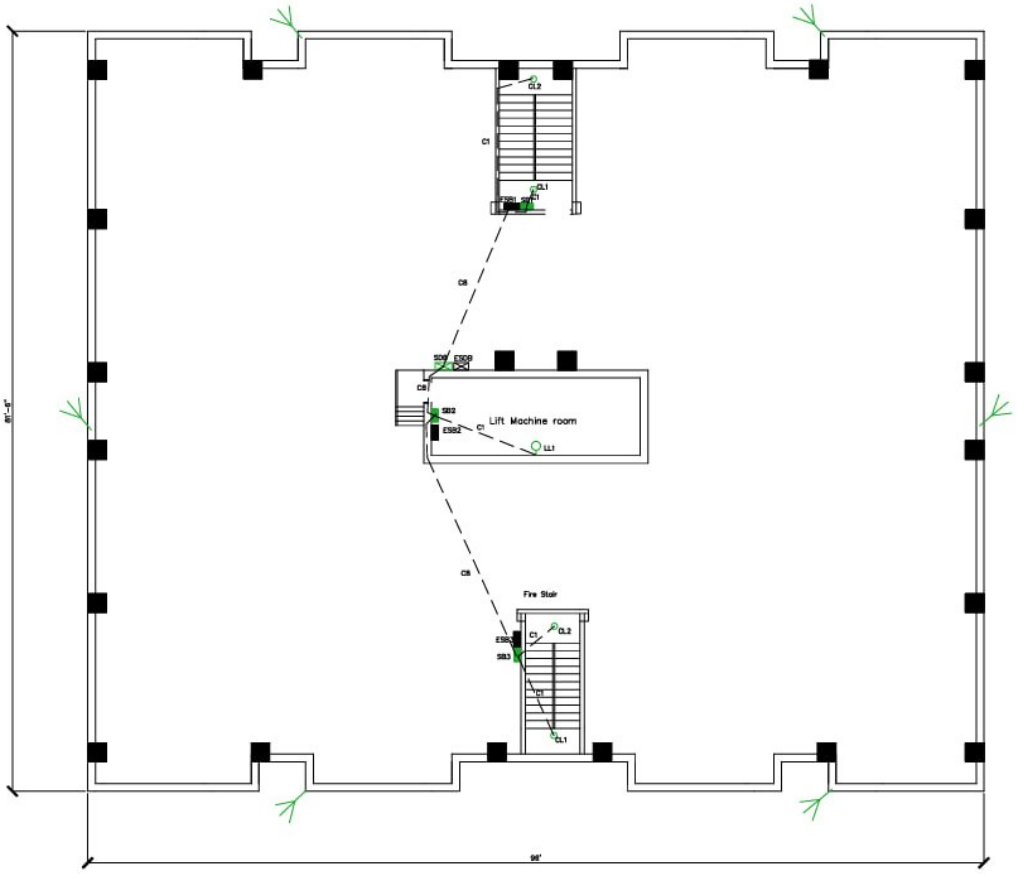


Figure: Rooftop — General Conduit Layout

Rooftop — Load / Current Calculation

ESDB-G current calculations

Table 4:ESDB-G current calculations

Circuit	Location / Description	Current	Total current (A)	Breaker to SDB
CKT1	Generator room (ESB2)	0.59 A	0.59	5 A
	Pump room (ESB2)			
CKT2	Main Stair (ESB3)	0.07 A	0.07	5 A
CKT3	Transformer room(ESB1)	0.52 A	0.59	5 A
	Fire Stair(ESB4)	0.07 A		
CTK 4	Driveway and others(ESB5)	0.14	0.14	5 A

Figure: Rooftop — Load and Current Calculation Table

Rooftop — Emergency Conduit Layout

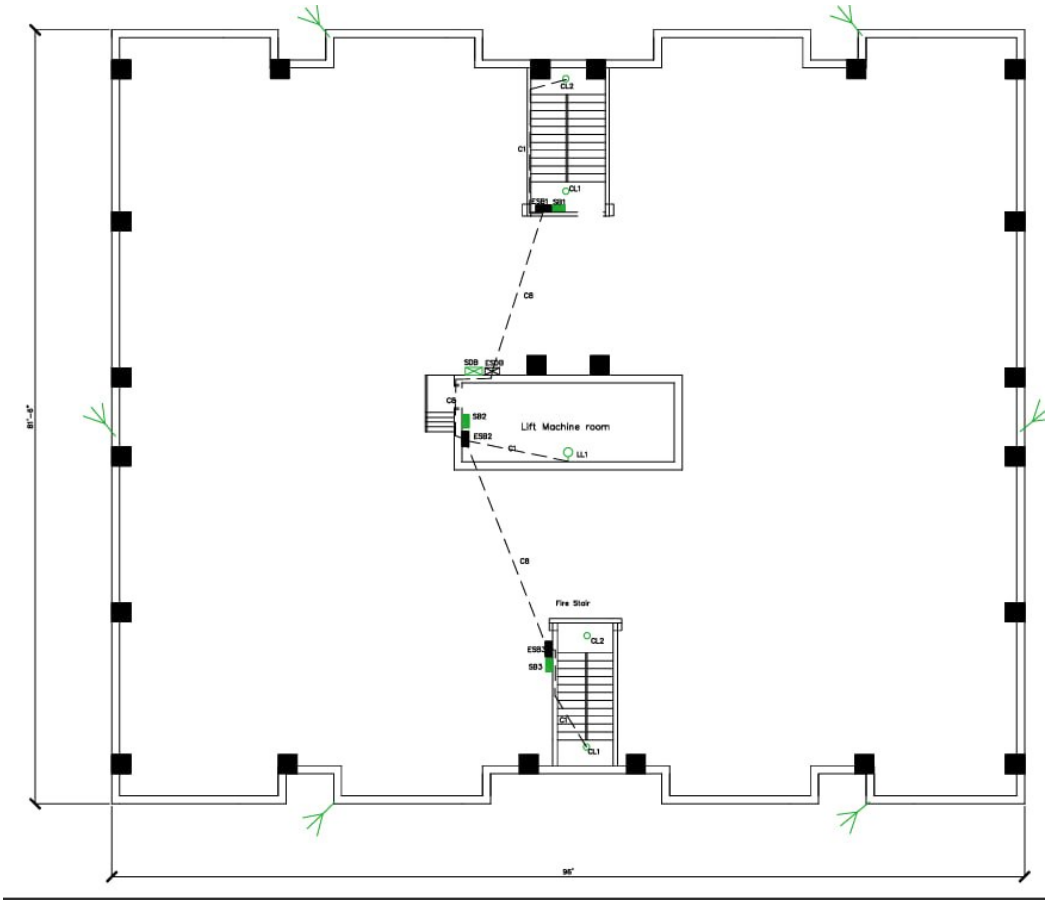


Figure: Rooftop — Emergency Conduit Layout

Rooftop — Emergency Load / Current Calculation

ESDB Roof load/current calculation

ESDB-Roof load Calculation								
Room name	Circuit no	Switch board	fixture	Power (W)	Current rating (A)	Wire rating	Conduit diagram	Breaker to SDB
Lift machine room	CKT1	ESB2	LL1	12	0.07	C8	20mm	5 A
Fire Stair		ESB3	CL1	12	0.07	C8	20mm	
Total current					0.14			
Main Stair	CKT2	ESB1	CL2	12	0.07	C8	20mm	5 A
Total current					0.07			

Figure: Rooftop — Emergency Load and Current Calculation

Rooftop — SDB Diagram

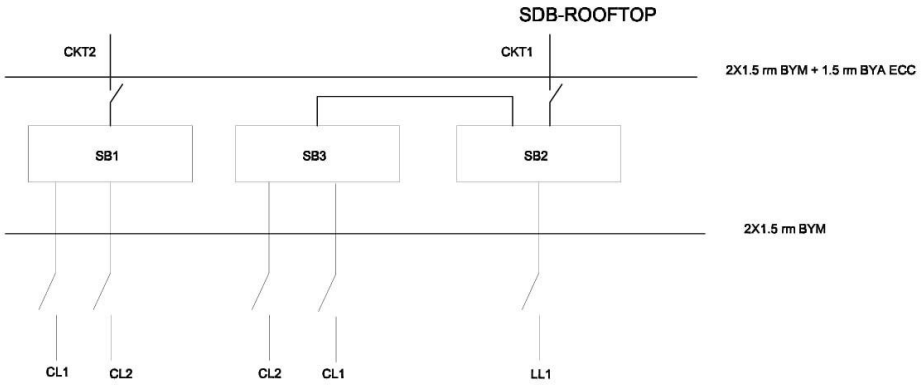


Figure: Rooftop SDB — Diagram

Rooftop — ESDB Diagram

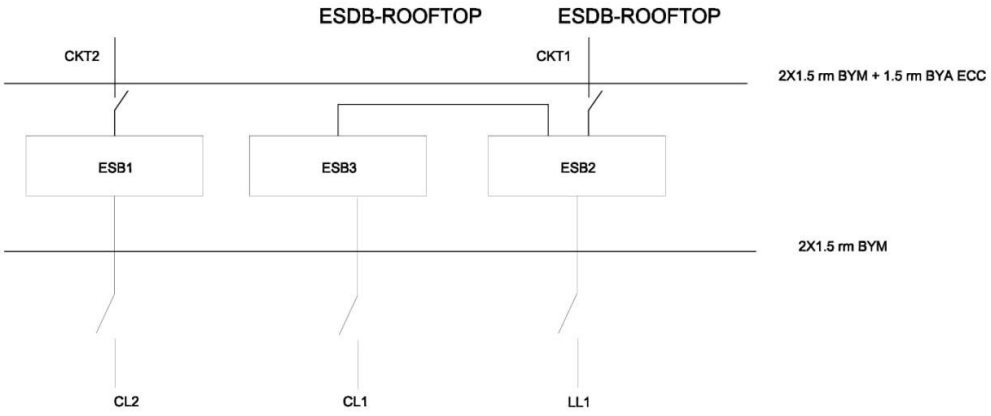


Figure: Rooftop ESDB Diagram

Rooftop — Lightning Protection System (LPS)

Table 1 - risk assessment

index	parameter	class	value
A	Use of structure	Houses and similar buildings	02
B	Type of construction	Brick, plain concrete or masonry with nonmetal roof	04
C	Contents or Consequential Effects	Ordinary domestic or office building, factories and workshops not containing valuable materials	02
D	Degree of Isolation	Located in a large area having structures or trees of similar or greater height	02
E	Type of Terrain	Flat terrain at any level	02
F	Height of Structure	46-53m	30
G	Lightning Prevalence	Over 21	21
total			63

Risk assessment factor is greater than 40. So **lightning protection is mandatory.**

LPS Design

Lightning arrestor

Let, the rod height = 2'
Roof perimeter = 355'
From rolling sphere method, $d_{max} = 2\sqrt{2rh - h^2}$
Here,
d = separation distance between the rods (meter)
r = radius of the rolling sphere (meter)
h = height of the rods (meter)
From NFPA-780, we got the following table:
Maximum values for rolling sphere radius corresponding to the class of LPS:

Class of LPS	Rolling sphere radius
I	20m
II	30m
III	45m
IV	60m

So, for 60m radius of the rolling sphere,
distance between two lightening arrestors, $d_{max} = 2\sqrt{2rh - h^2}$

$= 2\sqrt{2 * 60 * 0.6096 - (0.6096)^2}$

$= 17.06\ m$

= 56'

So, number of lightning arrestor $= \frac{355}{56} = 6.33 \approx \mathbf{6}$

- The arrestors shall be made with copper rods of minimum 12 mm diameter with tin coating on top.

Down Conductor

Total area = 7824 sq feet = 726.87 sq meter
Number of down conductor = 1 conductor for the 1st 80 sq m and $\left(\frac{726.87 - 80}{100}\right) = 6.46 \approx 7\ extra\ conductors.$

So, we need total of **8 down conductors**.
Earth termination resistance of ground electrodes – less than 10 ohms

Down conductor will be made of PVC insulated stranded annealed copper cable having 19 strands of 1.8 mm diameter

Roof Conductors

Roof conductors are placed 10” away from the roof railing connecting all the lightning arrestors to the down conductors. Roof conductor will be made of PVC insulated stranded annealed copper cable having 19 strands of 1.8 mm diameter Roof Layout with necessary lighting protection system is attached here.

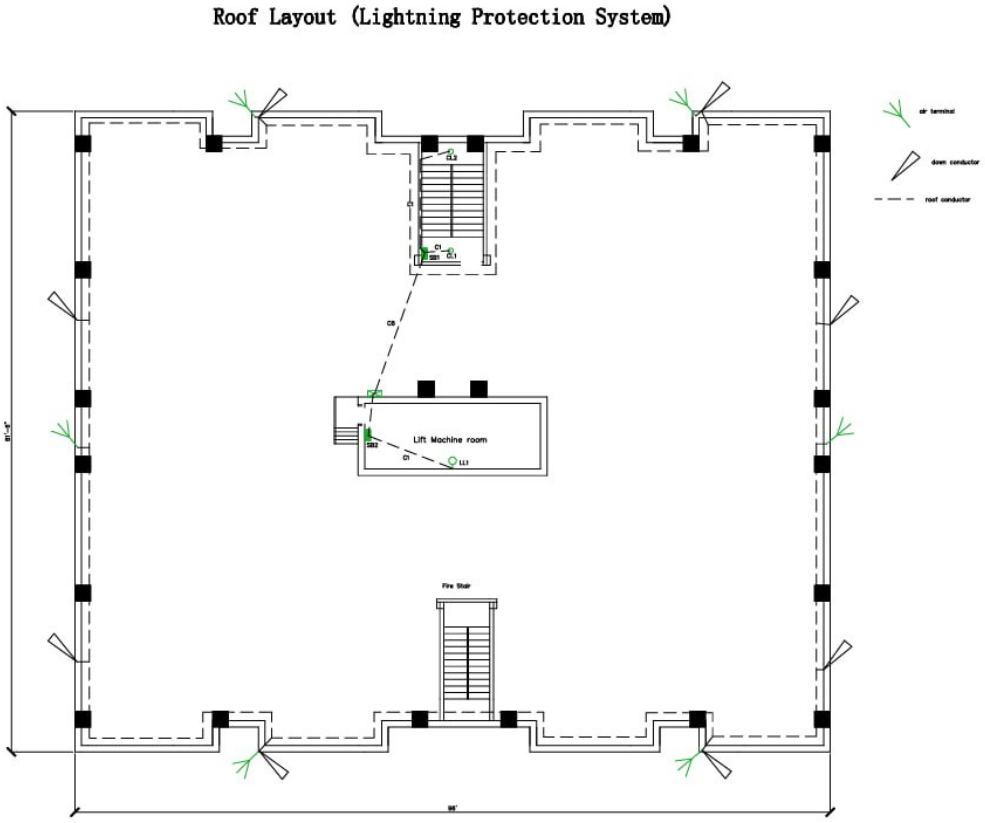


Figure: LPS — Down Conductor Layout

Basement Design

Basement Layout

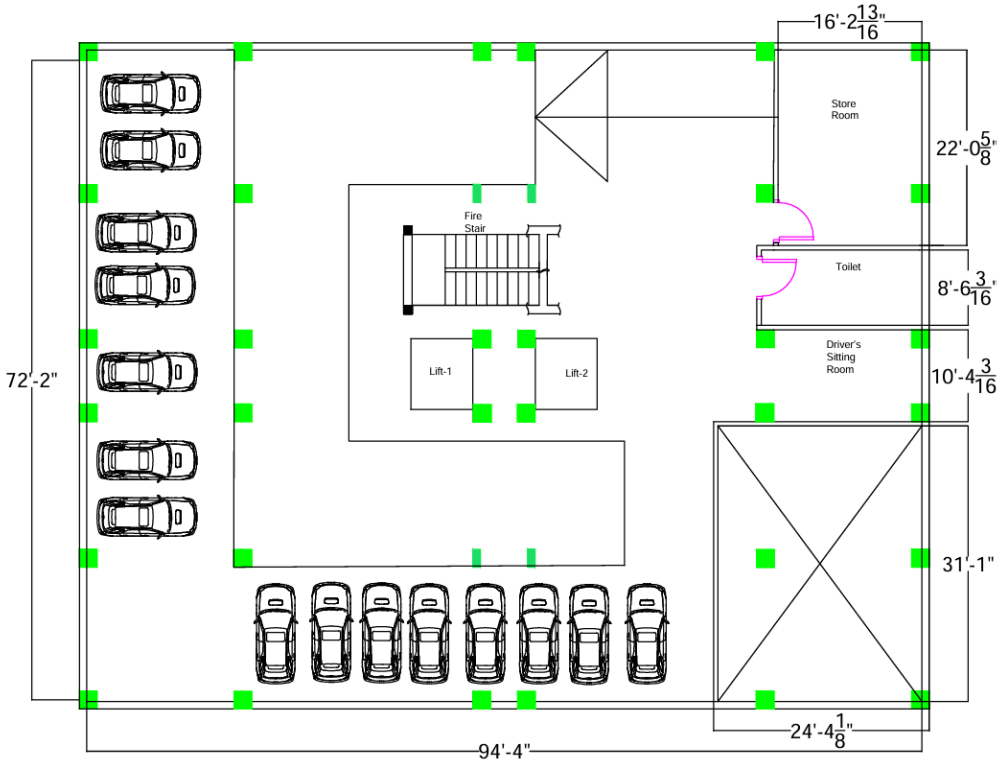


Figure: Basement Floor Layout

Basement Fixture Calculation

name of room	Illuminace(E)	n	lumen(F)	MF	Room index	UF	length	width	area	Light no(N)	Fan no (M)
Driver's Sitting Room	100	1	1250	0.8	1.098298193	0.62568	5.522913	3.154363	17.42127	2.784374188	1.875210157
Toilet	100	1	1250	0.8	0.965937605	0.61	5.522913	2.595563	14.33507	2.350011252	1.543013946
Store Room	70	1	1250	0.8	1.559117338	0.6736	6.721475	4.948238	33.25946	3.456297596	3.580018265
U.G.W.R	50	1	2250	0.8	2.276312119	0.705526	9.4742	7.419975	70.29833	2.767766616	7.566848958
Garages and Porches	50	1	2250	0.8	5.938670464	0.75	21.9964	21.43443	471.4803	17.46223319	50.7497167

Figure: Basement — Fixture Calculation Table

Basement Fittings and Fixtures

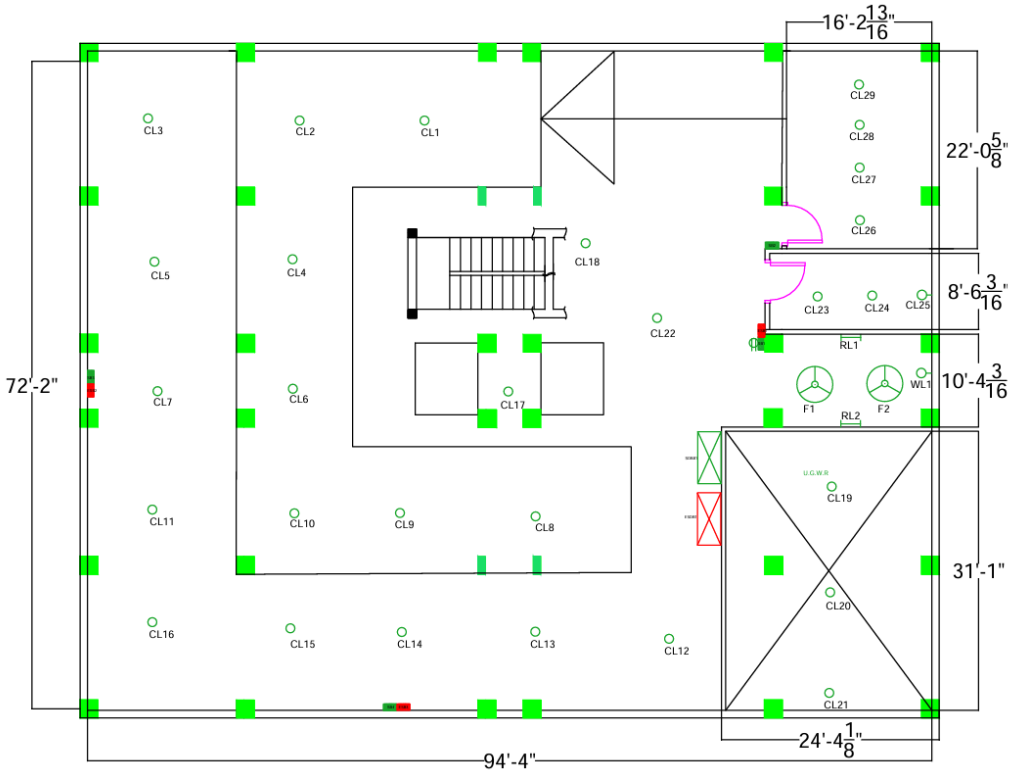


Figure: Basement — Fittings and Fixtures Layout

Basement — General Conduit Layout

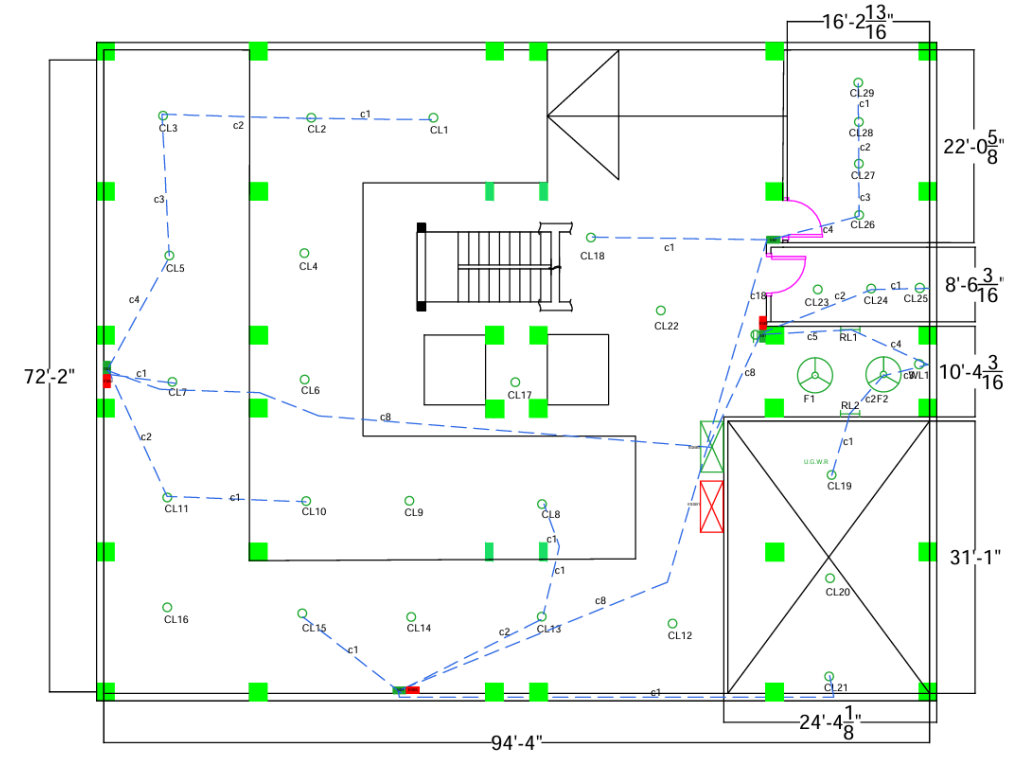


Figure: Basement — General Conduit Layout

Basement — SDB Diagram

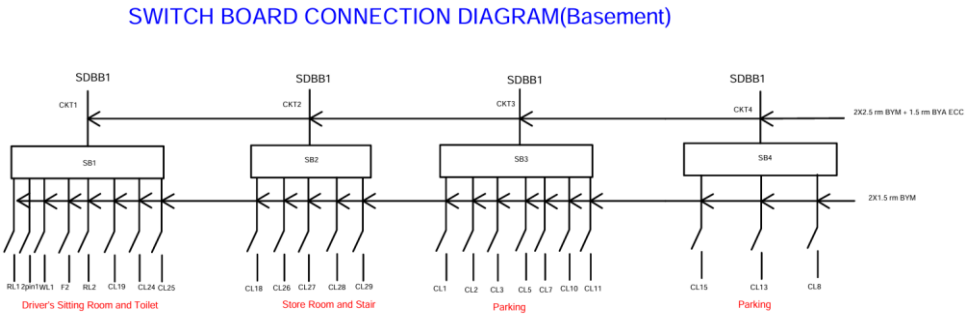


Figure: Basement SDB Diagram

SDB-B load calculations

Table 1 SDB-B load calculation

Room name	Circuit no	Switch board	fixture	Power (W)	Current rating (A)	Wire rating	Conduit diameter
Driver's Sitting Room and Toilet	CKT1	SB1	RL1 2pin1 WL1 F2 RL2 CL19 CL24 CL25	12 12 80 12 12 12 12	0.06 5 0.06 0.43 0.06 0.06 0.06 0.06	C8	20mm
Store Room and Stair	CKT2	SB2	CL18 CL26 CL27 CL28 CL29	12 12 12 12 12	0.06 0.06 0.06 0.06 0.06	C8	20mm
Parking	CKT3	SB3	CL1 CL2 CL3 CL5 CL7 CL10 CL11	12 12 12 12 12 12 12	0.06 0.06 0.06 0.06	C8	20mm
Parking	CKT4	SB4	CL15 CL13 CL18	12 12 12	0.06 0.06 0.06	C8	20mm

Figure: Basement SDB — Wire and CB Detail

SDB-B Current Calculations

Table 2: SDB-B Current calculation

Circuit	Location / Description	Current (A)	Total Current (A)	Breaker to SDB
CKT1	Driver's Sitting Room and Toilet (SB1)	RL1 — 12 W → 0.06 A	5.79	10 A
		2pin1 — 5 A (socket)		
		WL1 — 12 W → 0.06 A		
		F2 — 80 W → 0.43 A		
		RL2 — 12 W → 0.06 A		
		CL19 — 12 W → 0.06 A		
		CL24 — 12 W → 0.06 A		
		CL25 — 12 W → 0.06 A		
CKT2	Store Room and Stair (SB2)	CL18 — 12 W → 0.06 A	0.3	5 A
		CL26 — 12 W → 0.06 A		
		CL27 — 12 W → 0.06 A		
		CL28 — 12 W → 0.06 A		
		CL29 — 12 W → 0.06 A		
CKT3	Parking (SB3)	CL1 — 12 W → 0.06 A	0.42	5 A
		CL2 — 12 W → 0.06 A		
		CL3 — 12 W → 0.06 A		
		CL5 — 12 W → 0.06 A		
		CL7 — 12 W → 0.06 A		
		CL10 — 12 W → 0.06 A		
		CL11 — 12 W → 0.06 A		
CKT4	Parking (SB4)	CL15 — 12 W → 0.06 A	0.18	5 A
		CL13 — 12 W → 0.06 A		
		CL18 — 12 W → 0.06 A		

Figure: Basement SDB — Summary

Basement — Emergency Conduit Layout

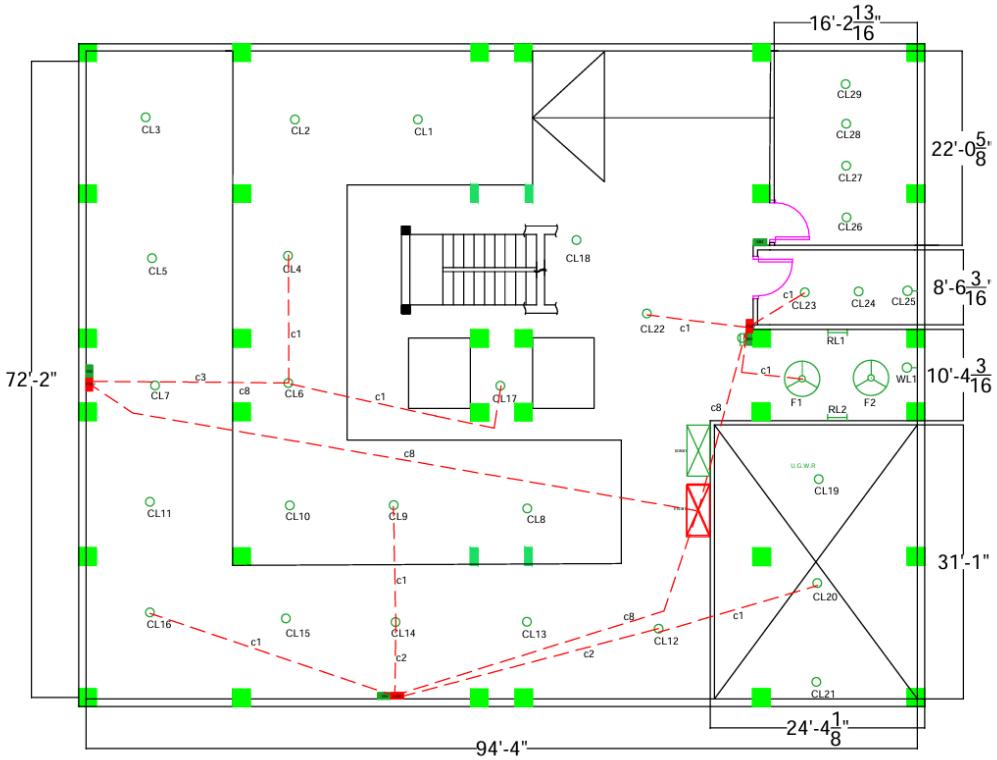


Figure: Basement — Emergency Conduit Layout

Basement — ESDB Diagram

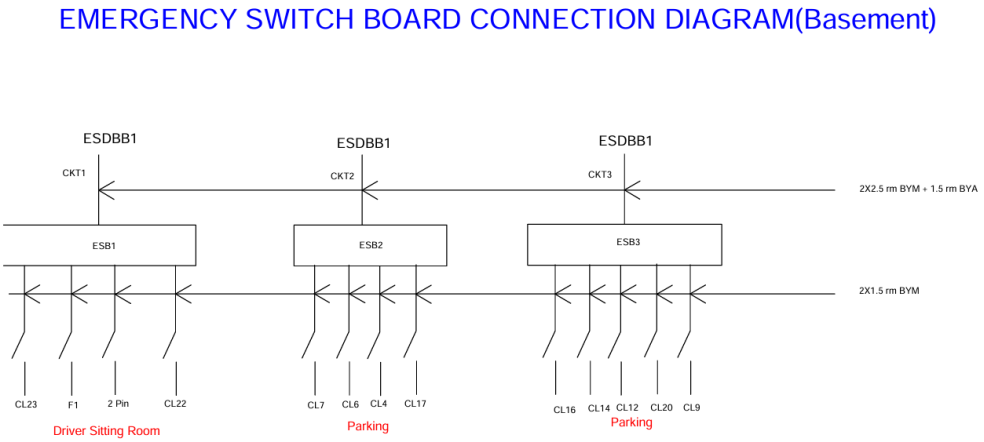


Figure: Basement ESDB Diagram

ESDB-B load calculations

Table 1 ESDB-B load calculation

Room name	Circuit no	Emergency Switch board	fixture	Power (W)	Current rating (A)	Wire rating	Conduit diameter
Drivers' Sitting Room and Toilet	CKT1	ESB1	CL 22 CL23 F1 2 Pin	12 12 80	0.06 0.06 0.43 5	C8	20mm
Parking	CKT2	ESB2	CL4 CL6 CL7 CL17	12 12 12 12	0.06 0.06 0.06 0.06	C8	20mm
Parking	CKT3	ESB3	CL9 CL12 CL14 CL16 CL20	12 12 12 12 12	0.06 0.06 0.06 0.06 0.06	C8	20mm

Figure: Earthing System — Connection Diagram

ESDB-B Current calculations

Table 2: ESDB-B Current calculation

Circuit	Location / Description	Current (A)	Total Current (A)	Breaker to SDB
CKT1	Driver's Sitting Room and Toilet (SB1)	CL 22 — 12 W → 0.06 A	5.55	10 A
		CL 22 — 12 W → 0.06 A		
		F1 — 80 W → 0.43 A		
		2pin1 — 5 A (socket)		
CKT2	Parking	CL4 — 12 W → 0.06 A	0.24	5 A
		CL6 — 12 W → 0.06 A		
		CL7 — 12 W → 0.06 A		
		CL17 — 12 W → 0.06 A		
CKT3	Parking	CL9 — 12 W → 0.06 A	0.3	5 A
		CL12 — 12 W → 0.06 A		
		CL14 — 12 W → 0.06 A		
		CL16 — 12 W → 0.06 A		
		CL20 — 12 W → 0.06 A		

Figure: ESDB Current Calculation

Earthing / Grounding System Design

Earthing Layout

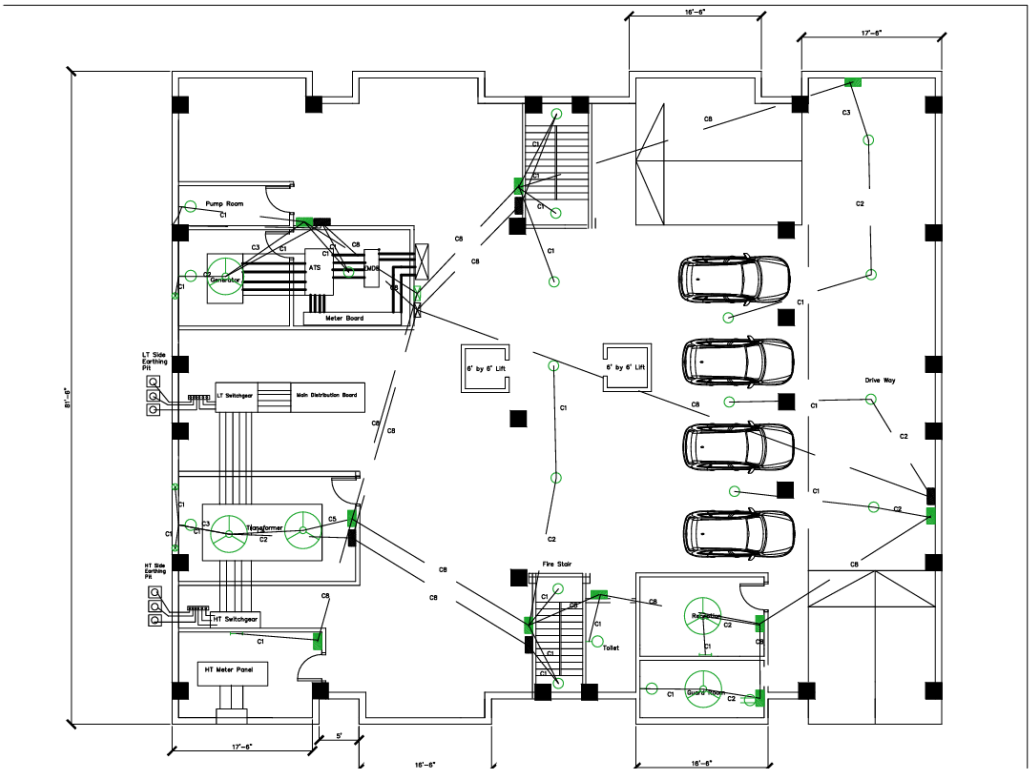


Figure: Earthing Layout — Ground Floor

Earth Pit Detail

GI pipe earth electrodes driven by tube well sinking method are recommended. Minimum pipe diameter is 38 mm; 50 mm is recommended for multi-storied buildings. Minimum pit internal dimensions: 600 mm × 600 mm × 600 mm. Alternate layers of charcoal and salt are used to reduce earth resistance to below 10 ohms.

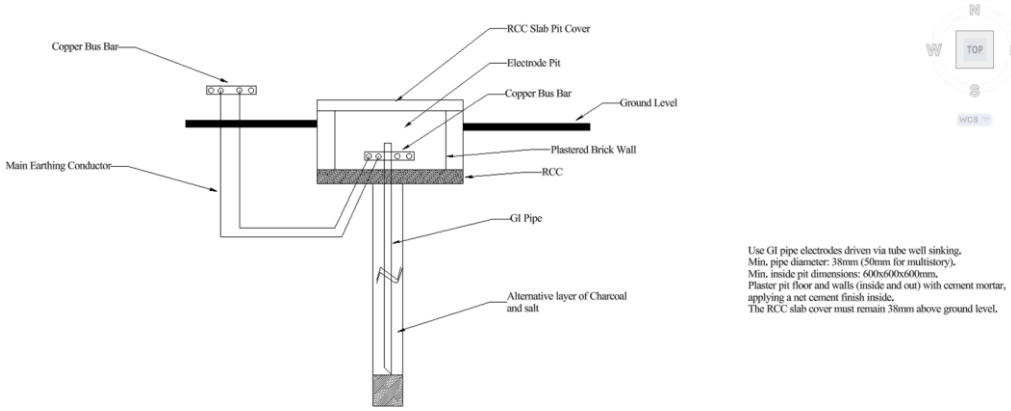


Figure: Earth Pit Construction Detail

Substation Design

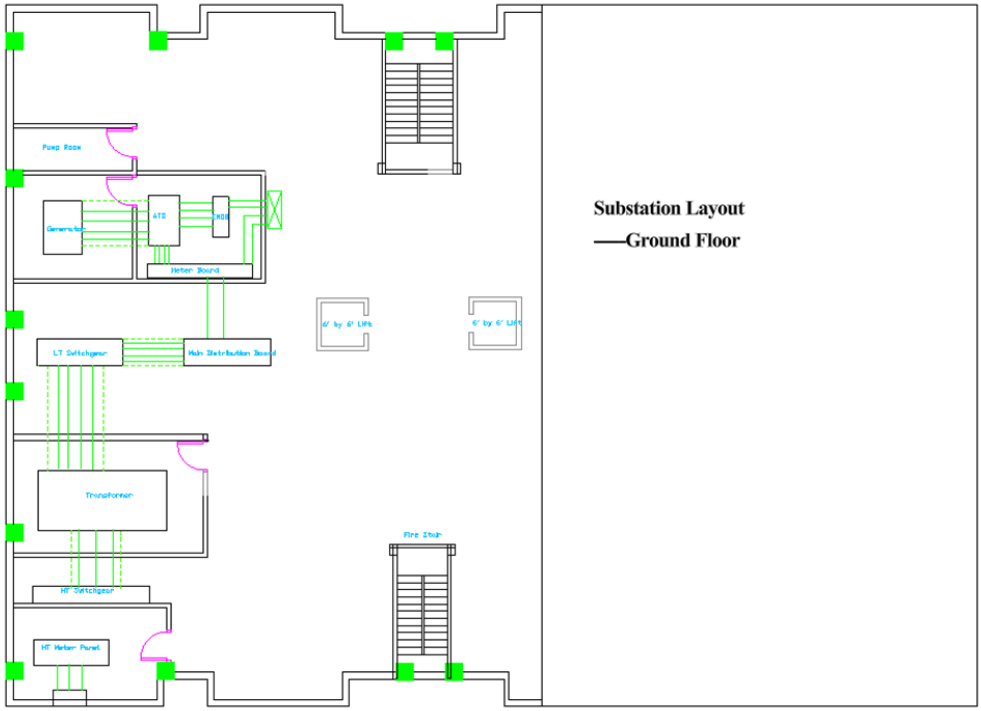


Figure: Substation Layout — Equipment Placement

Lift and Pump Calculations

Lift Calculation

Standard 14-floor buildings have a height of approximately 60 m. For this building, the car speed should be 1.6 m/s. Selected lift: 15 kW motor capacity, 10-person (800 kg) capacity, 2 units installed.

Table 8.4.9: Car Speed for Lift in Different Kinds of Usage

Building Type	Rise (m)	Minimum Car Speed (m/s)	Building Type	Rise (m)	Minimum Car Speed (m/s)
Office Building	0 to 40	2.0	Hospital (contd.)	31 to 40	1.6
	41 to 70	2.5		41 to 55	2.0
	71 to 85	3.6		56 to 75	2.5
	86 to 115	4.0		Above 75	3.6
	Above 115	5.0	Apartments	0 to 25	0.63
Hotels	0 to 40	2.0		26 to 40	1.0
	41 to 70	2.5		41 to 60	1.6
	71 to 85	3.6		Above 60	2.0
	86 to 115	4.0	Stores	0 to 30	1.0
	Above 115	5.0		31 to 45	1.6
Hospital	0 to 20	0.63		46 to 60	2.0
	21 to 30	1.0		Above 60	2.5

Figure: Lift Selection — BNBC Reference Table

Annexure-8

Table 8.1

Hoistway & Power Configurations for Passenger Lift According to Various Speed and Capacity

Persons	Rated Capacity (Kg) & maxm area(m2) as per EN 81	Rated Speed (m/s)	Net size of Car (Width x Depth) (mm)	Net size of Door (Width x Height) (mm)	Hoistway Size (Width x Depth) (mm)	Hoistway Size Overhead Height (mm)	Hoistway Size Pit Head (mm)	Motor Capacity (KW)	Machine Room Dimension (mm)	Maximum Current & |Running current| (Amp)
8	630 (1.66)	1.0	1200x1400	800x2100	1800x1900	4500	1500	4.0	1900x3700	14.8/9.2
		1.5/1.6				4600	1600	6.4		23.7/13.8
		2.0				4850	1800	8.3		30.9/17.1
		2.5				5100	1900	12.3		48.2/28.4
		3.0				5250	2100	14.1		57.3/32.8
		3.5				5400	2200	18.3		71/39
		4.0				5500	2300	24.0		96/52
10	800 (2.00)	1.0	1350x1450	900x2100	2000x2000	4500	1500	5.0	2100x3800	15.7/10.4
		1.5/1.6				4600	1600	8.1		25.8/16.2
		2.0				4850	1800	10.5		34.3/20.5
		2.5				5100	1900	15.0		55.7/34.4
		3.0				5250	2100	17.1		65.8/39.5
		3.5				5400	2200	22.7		86.4/46.3
		4.0				5500	2300	29.5		115.2/63

Figure: Lift Motor Power and Current Calculation

Pump and Lift Specification

Equipment	Rated Power	Running Current (per phase)	Selected Breaker (Inrush Protection)
Passenger Lift (x2)	15.0 kW (20 HP)	~25 A	100A TP MCCB
Water Pump	5.5 kW (7.5 HP)	~10 A	20A TP MCCB

Figure: Pump and Lift Specification Table

EMDB and Generator Calculation

Emergency Load Summary

The total emergency load is computed from all emergency distribution boards to size the generator and EMDB correctly.

Emergency Load Description	Phase Current	—
18 Units Emergency Lighting (~1.5 A per unit)	27.00 A	
Common Lifts (2 units × 25 A per phase)	50.00 A	
Common Water Pump (10 A running current per phase)	10.00 A	
ESDB-Ground & ESDB-Roof (1.39 A + 0.21 A)	1.60 A	
Total Emergency Phase Current (I_max_emergency)	88.60 A	Selected

Generator Sizing

Generator sizing calculation:

$$S_gen = (3 \times V_phase \times I_max_emergency) / 1000$$

$$S_gen = (3 \times 220\ V \times 88.6\ A) / 1000 = 58.4\ kVA$$

Generator Rating Selected: 100 kVA, 415 V, 3-Phase

EMDB Main Incomer Breaker: 125A TP MCCB

ATS (Automatic Transfer Switch) Rating: 125A

Emergency Load Description	Phase Current
18 Units Emergency Lighting (~1.5 A per unit)	27.00 A
Common Lifts (2 units x 25 A per phase)	50.00 A
Common Water Pump (10 A running current per phase)	10.00 A
ESDB-Ground & ESDB-Roof (1.39 A + 0.21 A)	1.60 A
Total Emergency Phase Current	88.60 A

Figure: EMDB and Generator Calculation Summary

EMDB Single Line Diagram

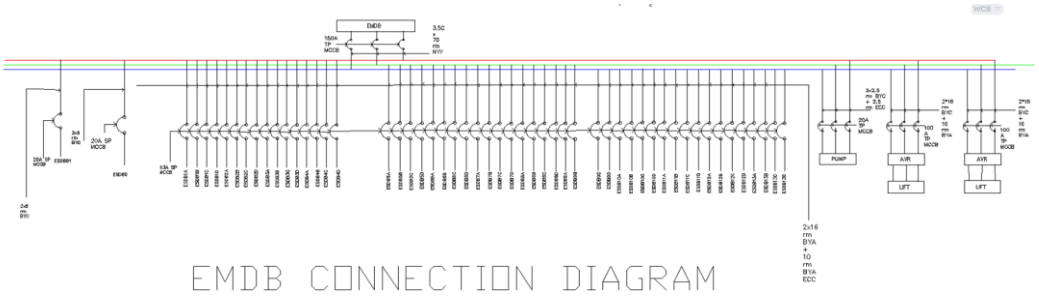


Figure: Single Line Diagram — EMDB

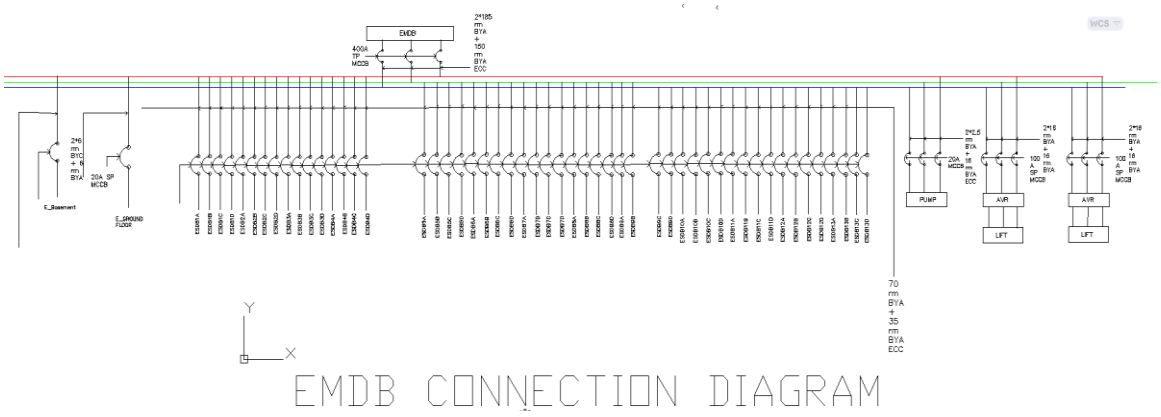


Figure: EMDB SLD — Substation Reference Diagram

MDB, Transformer and PFI Plant Calculation

MDB and Transformer Calculation

Total maximum current on the heaviest phase is calculated from all loads:

Load Description	Phase Current	—
18 Typical Units (18 × 62.46 A)	1124.28 A	
Common Lifts (2 units × 25 A per phase)	50.00 A	
Common Water Pump (10 A running current per phase)	10.00 A	
SDB-Ground (from load schedule)	8.49 A	
SDB-Roof (from load schedule)	0.35 A	
Total Phase Current (I_max)	1193.12 A	Selected

Transformer Sizing

Using the standard 3-phase apparent power equation:

$$S = 3 \times V_phase \times I_max$$

$$S = 3 \times 220\ V \times 1193.12\ A = 787{,}459\ VA = 787.46\ kVA$$

Transformer Rating Selected: 1000 kVA, 11 kV/0.415 kV, 3-Phase, Delta-Wye
MDB Main Incomer Breaker: 1250A ACB (Air Circuit Breaker)
HT Switchgear Rating: 630A, 11 kV VCB (Vacuum Circuit Breaker)

Load Description	Phase Current
18 Typical Units (18 x 62.46 A)	1124.28 A
Common Lifts (2 units x 25 A per phase)	50.00 A
Common Water Pump (10 A running current per phase)	10.00 A
SDB-Ground (From load schedule)	8.49 A
SDB-Roof (From load schedule)	0.35 A
Total Phase Current (I_max)	1193.12 A

Figure: MDB and Transformer Calculation Summary

MDB Single Line Diagram

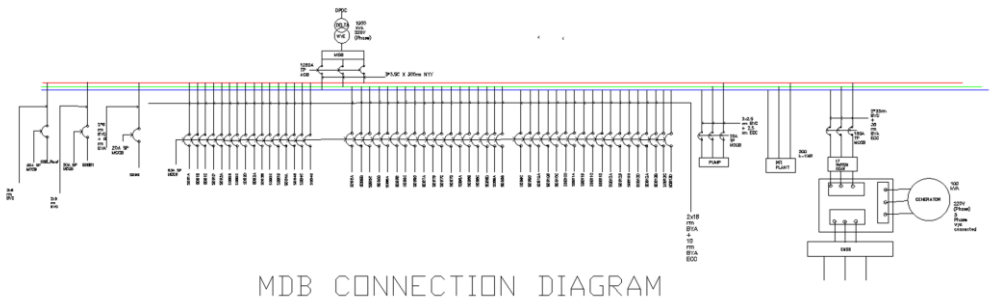


Figure: Single Line Diagram — MDB

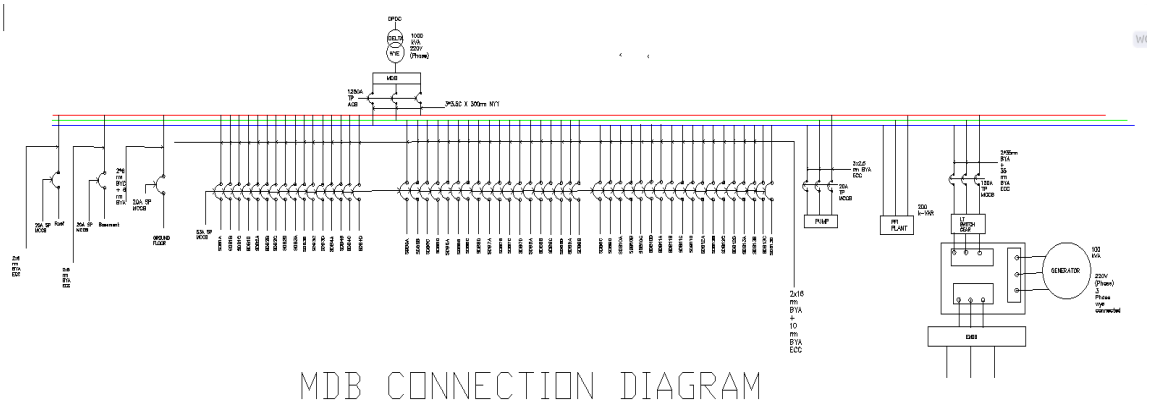


Figure: MDB SLD — Substation Reference Diagram

Table 8.4.9: Car Speed for Lift in Different Kinds of Usage

Building Type	Rise (m)	Minimum Car Speed (m/s)	Building Type	Rise (m)	Minimum Car Speed (m/s)
Office Building	0 to 40	2.0	Hospital (contd.)	31 to 40	1.6
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	Above 115	5.0	Apartments	0 to 25	0.63
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	86 to 115	4.0	Stores	0 to 30	1.0
	Above 115	5.0		31 to 45	1.6
Hospital	0 to 20	0.63		46 to 60	2.0
	21 to 30	1.0		Above 60	2.5

Figure: Substation Equipment Sizing — Calculation Tables

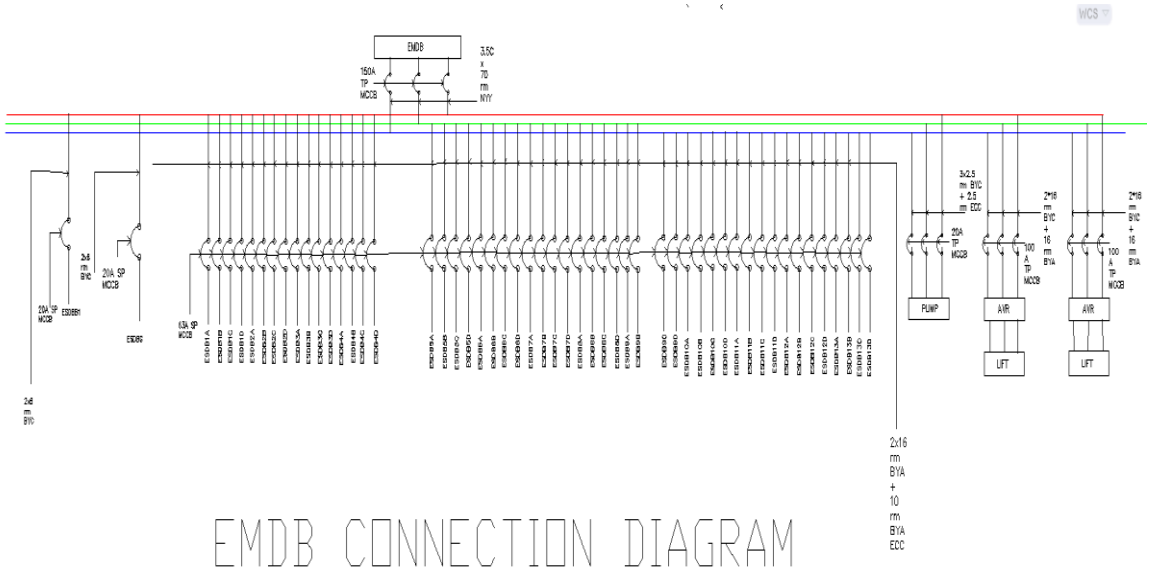


Figure: Substation Sizing — Cable and PFI Schedule

PFI Plant Calculation

To comply with standard grid regulations, the Power Factor must be improved from 0.8 to a target of 0.9.

Real Power (P) Calculation:

$$P = (3 \times V_phase \times I_max \times cos\Phi_target) / 1000$$

$$P = (3 \times 220 \times 1193.12 \times 0.9) / 1000 = 708.7\ kW$$

Reactive Power Calculation:

$$Q_worst\ (at\ 0.8\ PF) = P \times \sqrt{((1/0.8)^2 - 1)} = 708.7 \times 0.75 = 531.5\ kVAR$$

$$Q_best\ \ (at\ 0.9\ PF) = P \times \sqrt{((1/0.9)^2 - 1)} = 708.7 \times 0.484 = 343.0\ kVAR$$

Required PFI Rating:

$$Q_req = Q_worst - Q_best = 531.5 - 343.0 = 188.5\ kVAR$$

PFI Plant Selected: 200 kVAR Automatic PFI Plant

Conclusion

In this project, the complete electrical service design of a 4-unit, 14-storied residential building has been carried out. Starting from the unit floor layout, fixture and conduit calculations were performed for the typical unit, ground floor, basement, and rooftop. Switchboard and distribution board diagrams from the switchboard level up to the Main Distribution Board (MDB) were developed. A substation with a 1000 kVA transformer and a 100 kVA emergency generator with ATS were sized and designed. The earthing system was laid out and earth pit details were specified. A Lightning Protection System was designed using the rolling sphere method in accordance with NFPA-780 and BNBC 2020. Through this project, comprehensive hands-on experience in the design of electrical services for a large residential building was gained.

References

1. Bangladesh National Building Code (BNBC) 2020
2. Public Works Department (PWD) Document
3. Lecture slides and materials provided in EEE 414 course
4. NFPA 780 — Standard for the Installation of Lightning Protection Systems